03

时序逻辑电路



3 - 1 锁存器、触发器

3 - 2 计数器

3 - 3 时序逻辑电路设计

一、触发器 Flip-Flop

能够存储1位二值信号的基本单元电路

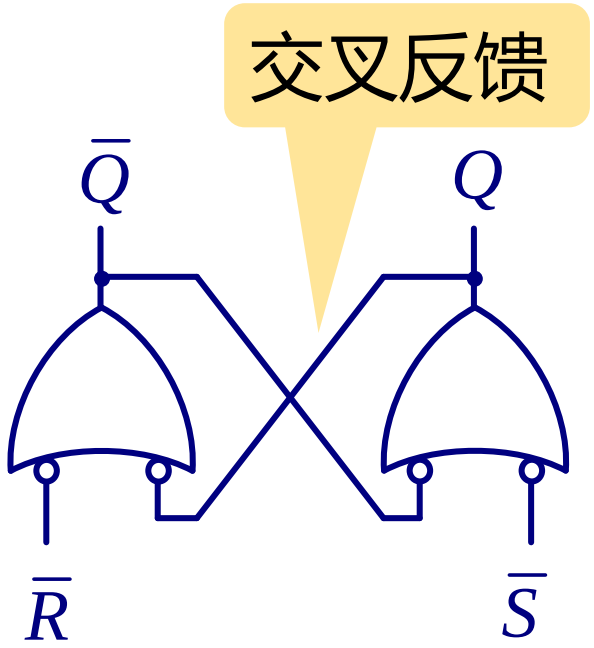
触发器的功能：形象地说，它具有“一触即发”的功能。在输入信号的作用下，输出能够从一种状态（0或1）转变成另一种状态（1或0）。

触发器的特点：有记忆功能的逻辑部件。输出状态不只与当前的输入有关，还与原来的输出状态有关。

触发器分类

- 按工作状态分
  - 双稳态触发器 Bistable
  - 单稳态触发器 Monostable
  - 无稳态触发器 Astable
- 按逻辑功能分
  - SR触发器
  - JK触发器
  - D触发器
  - T触发器
- 按触发方式分
  - 基本锁存器
  - 门控锁存器
  - 主从触发器
  - 边沿触发器

1、基本 S-R锁存器 The Set-Reset Latch



Active-LOW input

正常情况下，两输出互补

$Q=1$ $\bar{Q}=0$ 称为 “1” 态

$Q=0$ $\bar{Q}=1$ 称为 “0” 态

特殊情况下，两输出相同

$Q=1$ $\bar{Q}=1$ 记作 1*

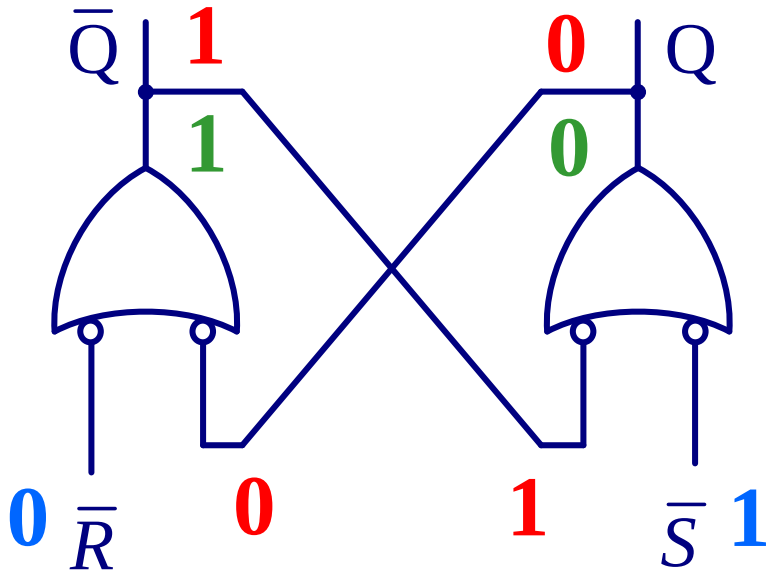
$Q=0$ $\bar{Q}=0$ 记作 0*

正是由于引入反馈，才使电路具有记忆功能！

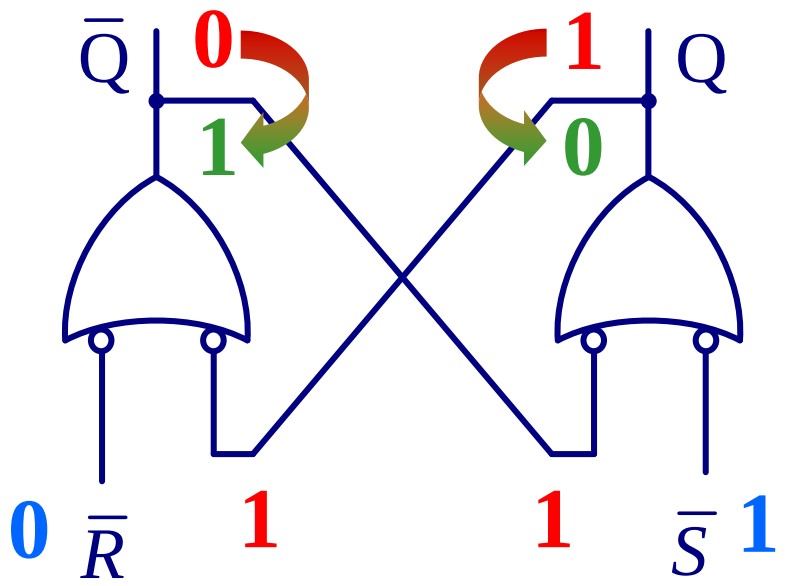
输入$\overline{R}$=0, $\overline{S}$=1时

置“0”！ **Reset**

若原状态：$Q^n=0$ $\bar{Q}^n=1$



若原状态：$Q^n=1$ $\bar{Q}^{n+1}=0$



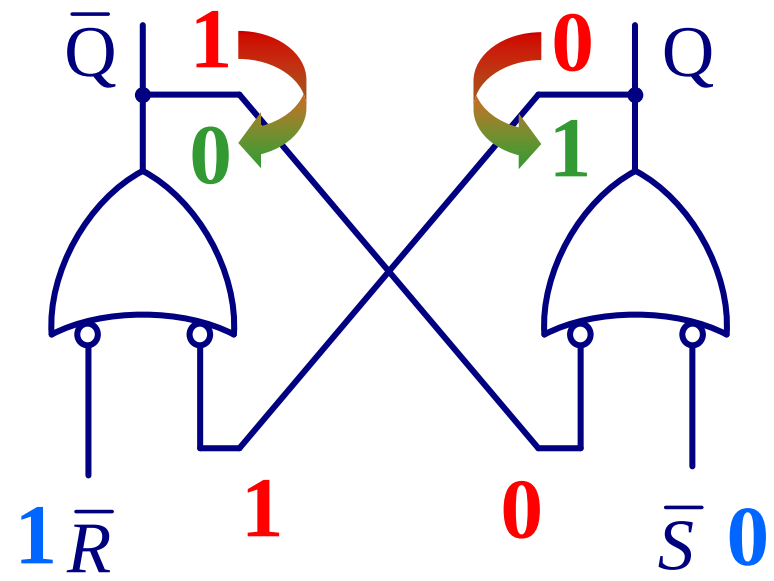
输出仍保持：$Q^{n+1}=0$ $\bar{Q}^{n+1}=1$

输出变为：$Q^{n+1}=0$ $\bar{Q}^{n+1}=1$

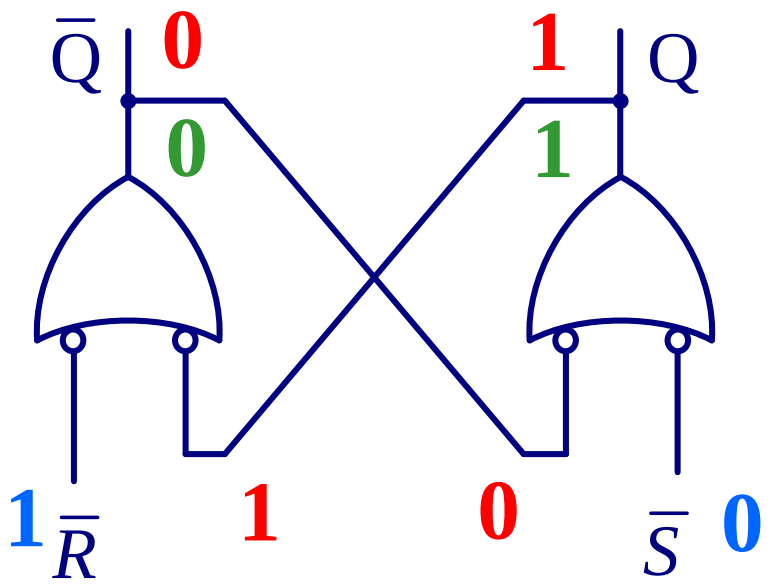
输入$\overline{R}$=1, $\overline{S}$=0 时

置 "1"　Set

若原状态：$Q^n=0$　$\bar{Q}^n=1$



若原状态：$Q^n=1$　$\bar{Q}^{n+1}=0$



输出变为：　$Q^{n+1}=1$　$\bar{Q}^{n+1}=0$

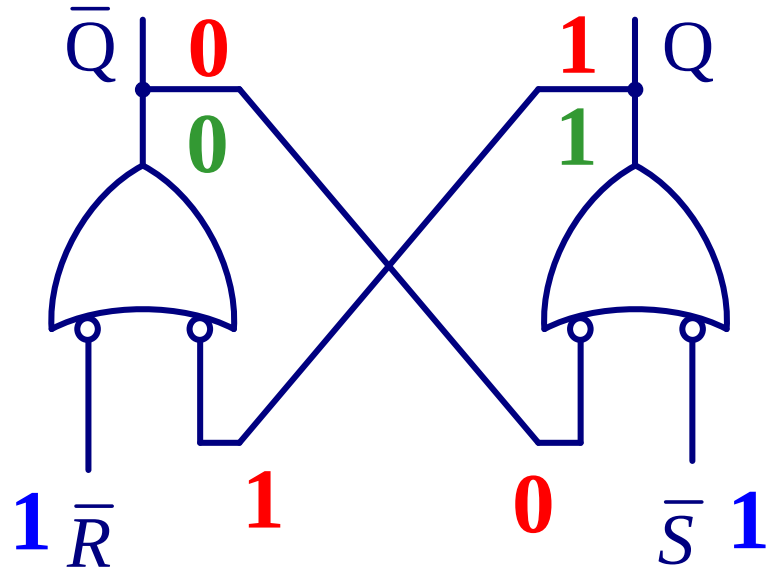
输出保持：　$Q^{n+1}=1$　$\bar{Q}^{n+1}=0$

输入$\overline{R}$=1, $\overline{S}$=1时

保持!

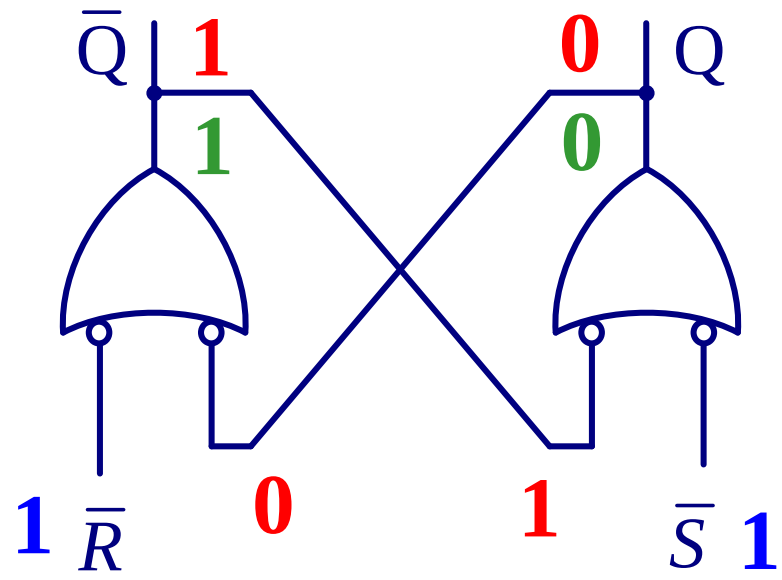
$Q^{n+1} = Q^n$

若原状态：$Q^n = 1 \quad \overline{Q}^{n+1} = 0$



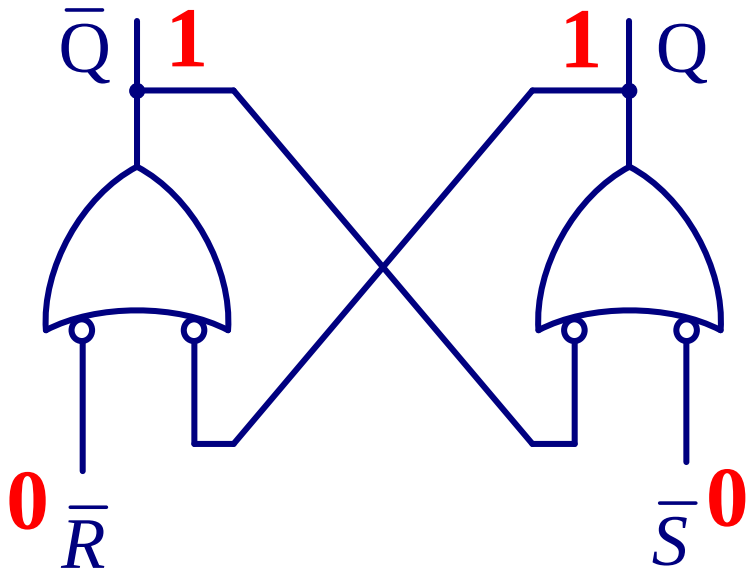
输出保持原状态：$Q^{n+1} = 1 \quad \overline{Q}^{n+1} = 0$

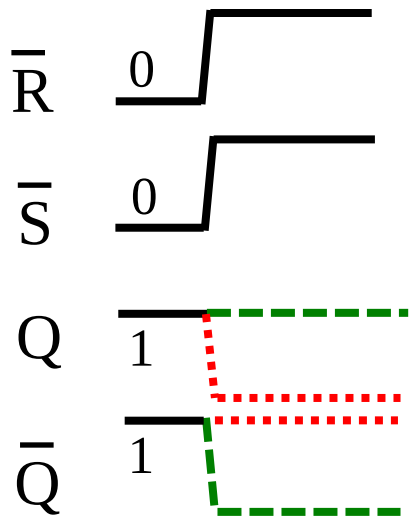
若原状态：$Q^n = 0 \quad \overline{Q}^n = 1$



输出保持原状态：$Q^{n+1} = 0 \quad \overline{Q}^{n+1} = 1$

输入$\overline{R}$=0, $\overline{S}$=0时　　输出：全是1





$\overline{R}=0\ \ \overline{S}=0\Rightarrow Q^n=1\ \ \overline{Q}^n=1$

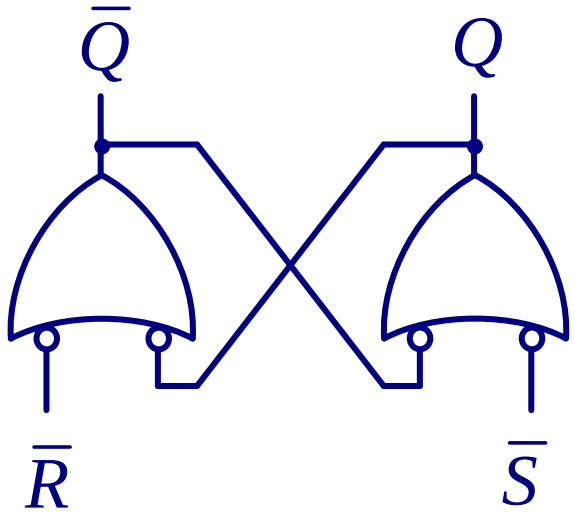
$(1)\overline{R}=0\ \ \overline{S}=1\Rightarrow Q^{n+1}=0\ \ \overline{Q}^{n+1}=1$

$(2)\overline{R}=1\ \overline{S}=0\Rightarrow Q^{n+1}=1\ \ \overline{Q}^{n+1}=0$

$(3)\overline{R}=1\ \ \overline{S}=1\Rightarrow$ 不定

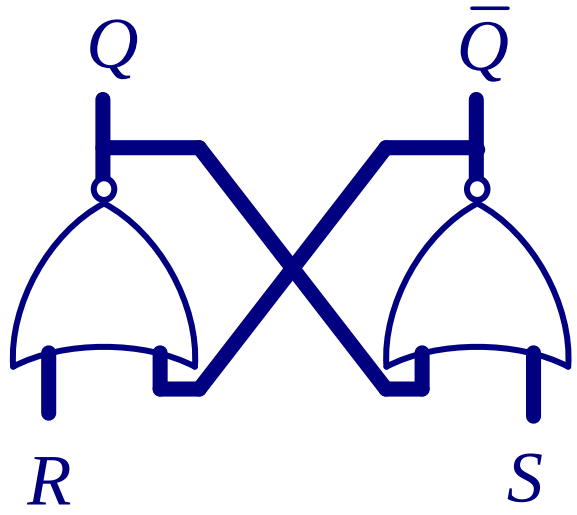
$(4)\overline{R}=0\ \ \overline{S}=0\Rightarrow Q^{n+1}=1\ \ \overline{Q}^{n+1}=1$

注意：当$\overline{R}$、$\overline{S}$同时由0变为1时，翻转快的门输出变为0，另一个不得翻转。因此，该状态不定



$\bar{R}$	$\bar{S}$	Q^n	$\bar{Q}^n$	Q^{n+1}	$\bar{Q}^{n+1}$	
0	0	0	1	1	1	1^*
0	0	1	0	1	1	
0	1	0	1	0	1	0
0	1	1	0	0	1	
1	0	0	1	1	0	1
1	0	1	0	1	0	
1	1	0	1	0	1	Q^n
1	1	1	0	1	0	

$\bar{R}$	$\bar{S}$	Q^{n+1}	
0	0	1^*	不定
0	1	0	复位
1	0	1	置位
1	1	Q^n	保持



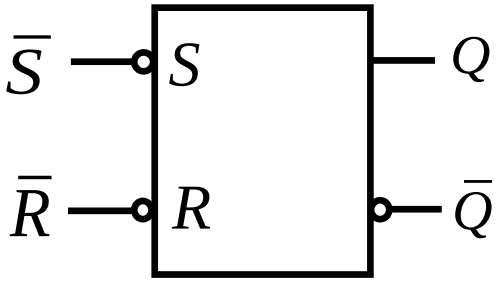
R	S	Q^n	$\bar{Q}^n$	Q^{n+1}	$\bar{Q}^{n+1}$	
0	0	0	1	0	1	Q^n
0	0	1	0	1	0	
0	1	0	1	1	0	1
0	1	1	0	1	0	
1	0	0	1	0	1	0
1	0	1	0	0	1	
1	1	0	1	0	0	0^*
1	1	1	0	0	0	

禁用

R	S	Q^{n+1}	
0	0	Q^n	保持
0	1	1	置位
1	0	0	复位
1	1	0^*	不定

基本S-R锁存器的特性方程

逻辑符号

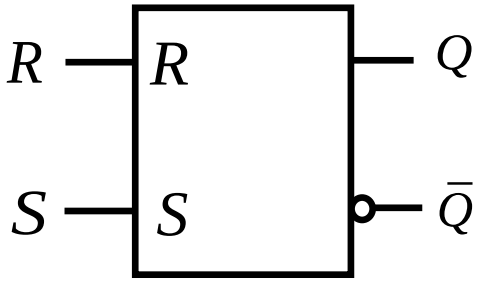


Q^{n+1} / $\overline{S}$ $\overline{R}$	00	01	11	10
Q^n = 0	×	1	0	0
Q^n = 1	×	1	1	0

特性方程

$$\begin{cases} Q^{n+1} = \overline{S} + \overline{R}Q^n \\ \overline{S} + \overline{R} = 1 \end{cases}$$

约束条件



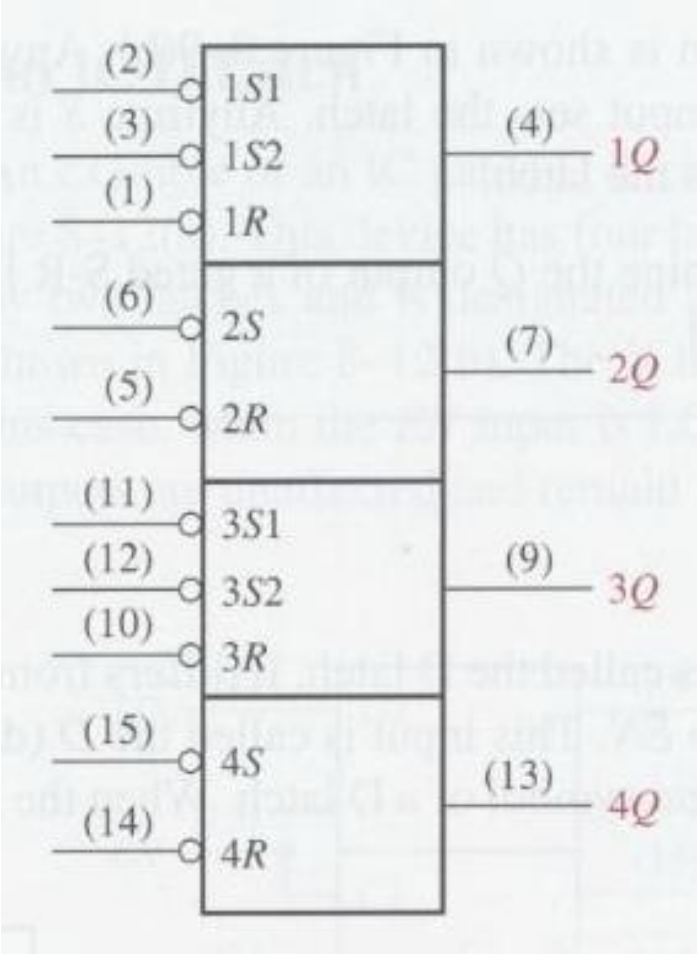
Q^{n+1} / S R	00	01	11	10
Q^n = 0	0	0	×	1
Q^n = 1	1	0	×	1

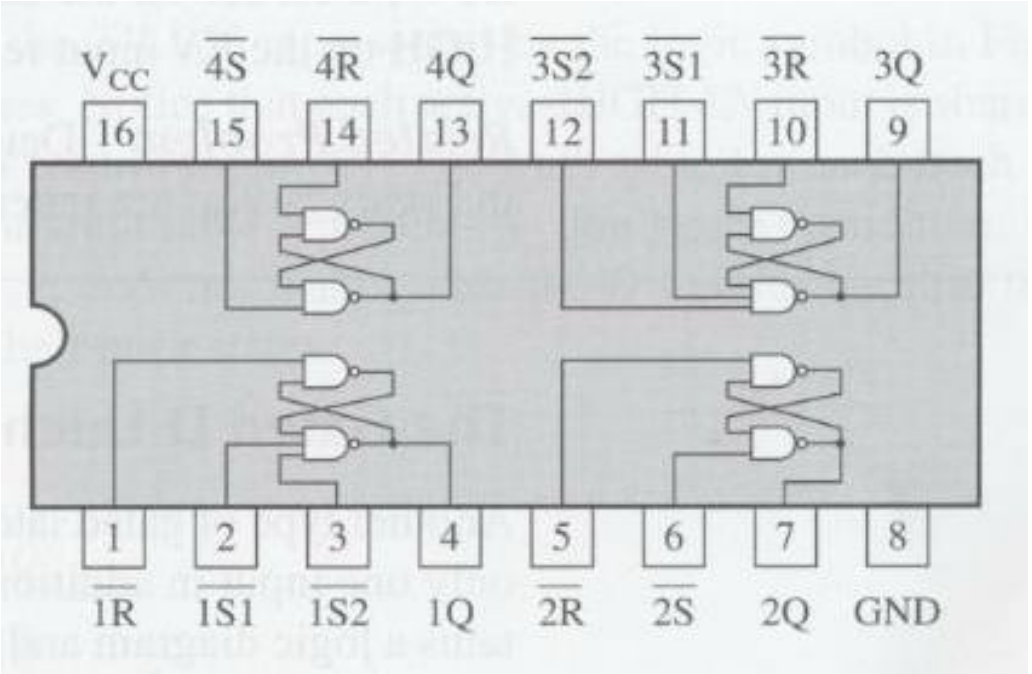
$$\begin{cases} Q^{n+1} = S + \overline{R}Q^n \\ S \cdot R = 0 \end{cases}$$

约束条件

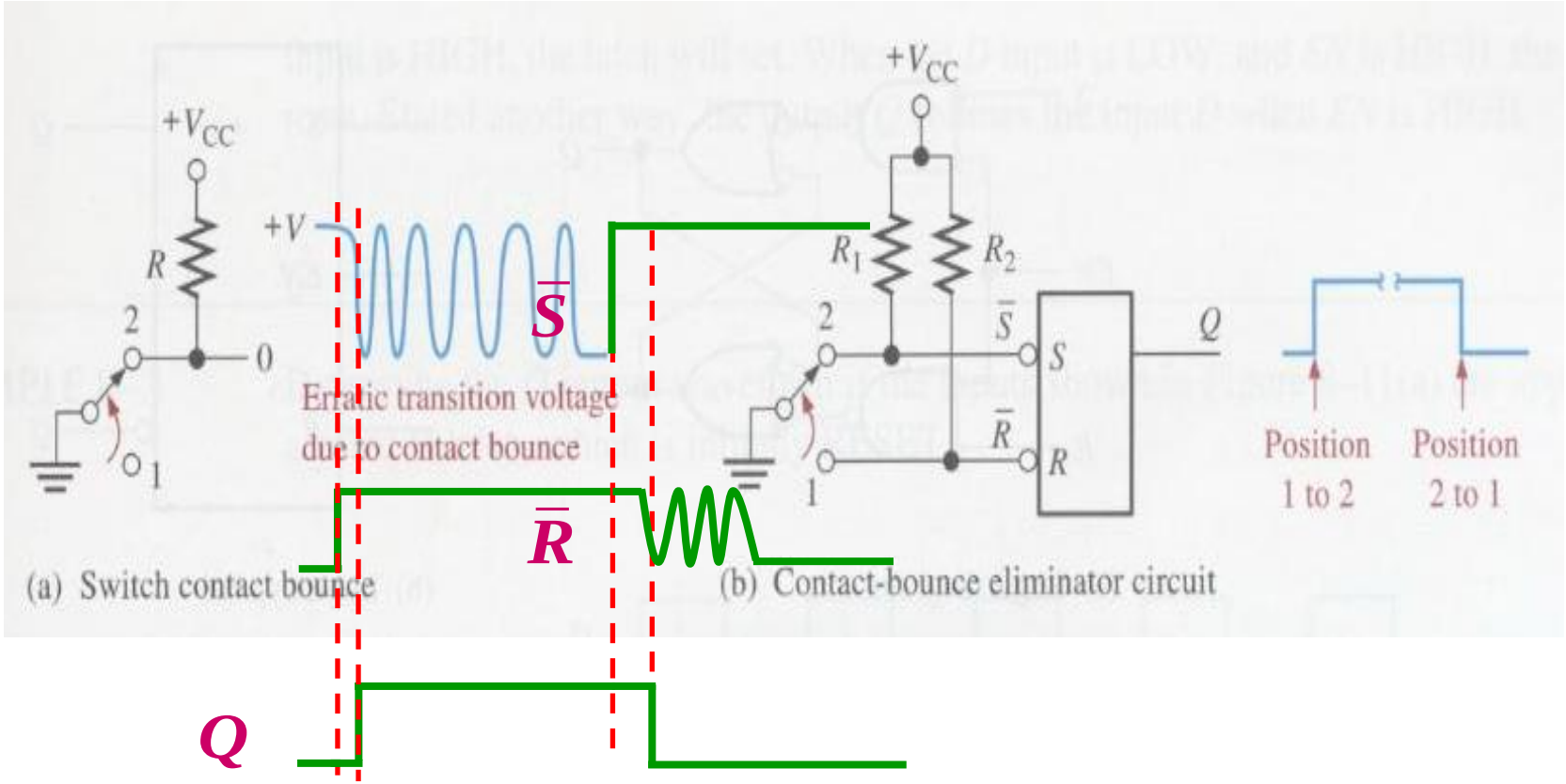
S J T U

74LS279　四重$\overline{S}$-$\overline{R}$锁存器

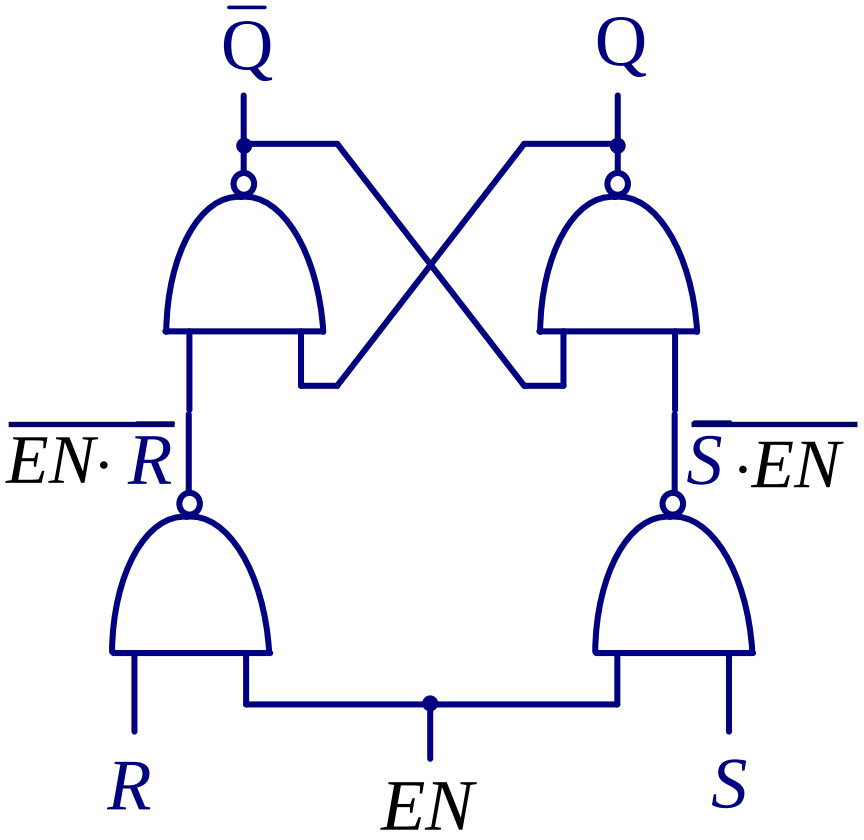




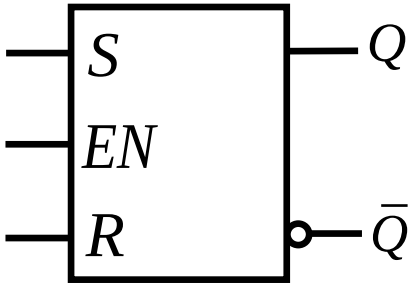
应用：消除机械开关的接触抖动



2、门控S-R锁存器 The gated S-R Latch



逻辑符号



基本 S-R锁存器

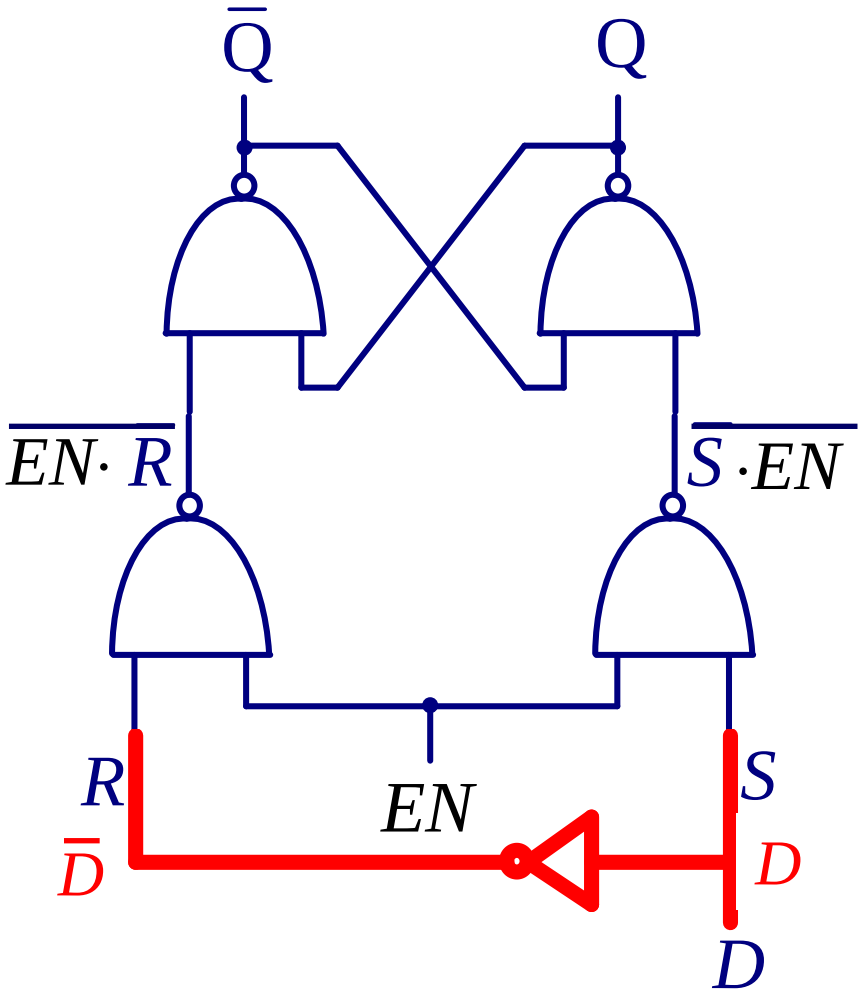
R	S	Q^{n+1}
0	0	Q^n
0	1	1
1	0	0
1	1	0^*

门控S-R锁存器特性表

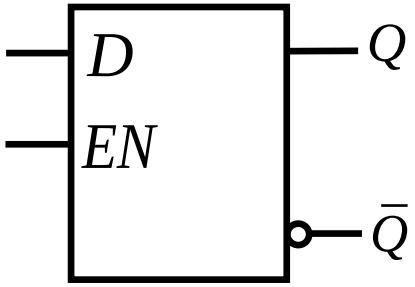
EN	R	S	Q^{n+1}
0	×	×	Q^n
1	0	0	Q^n
1	0	1	1
1	1	0	0
1	1	1	1*

$$\begin{cases} Q^{n+1} = S + \bar{R}Q^n \\ SR = 0 \end{cases}$$ 约束条件

3、门控D锁存器 The gated D Latch



逻辑符号

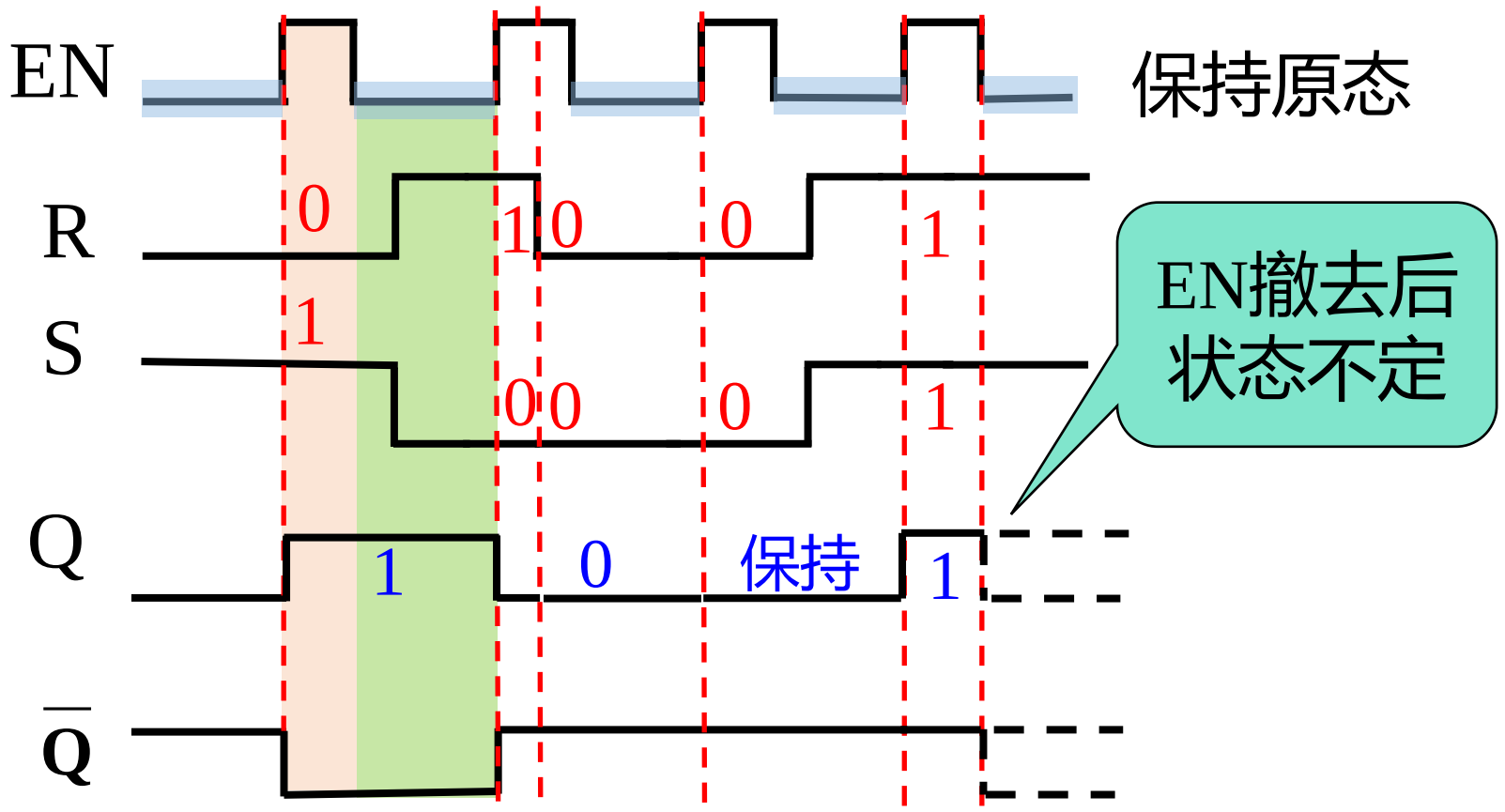


$$Q^{n+1} = D$$

门控D锁存器特性表

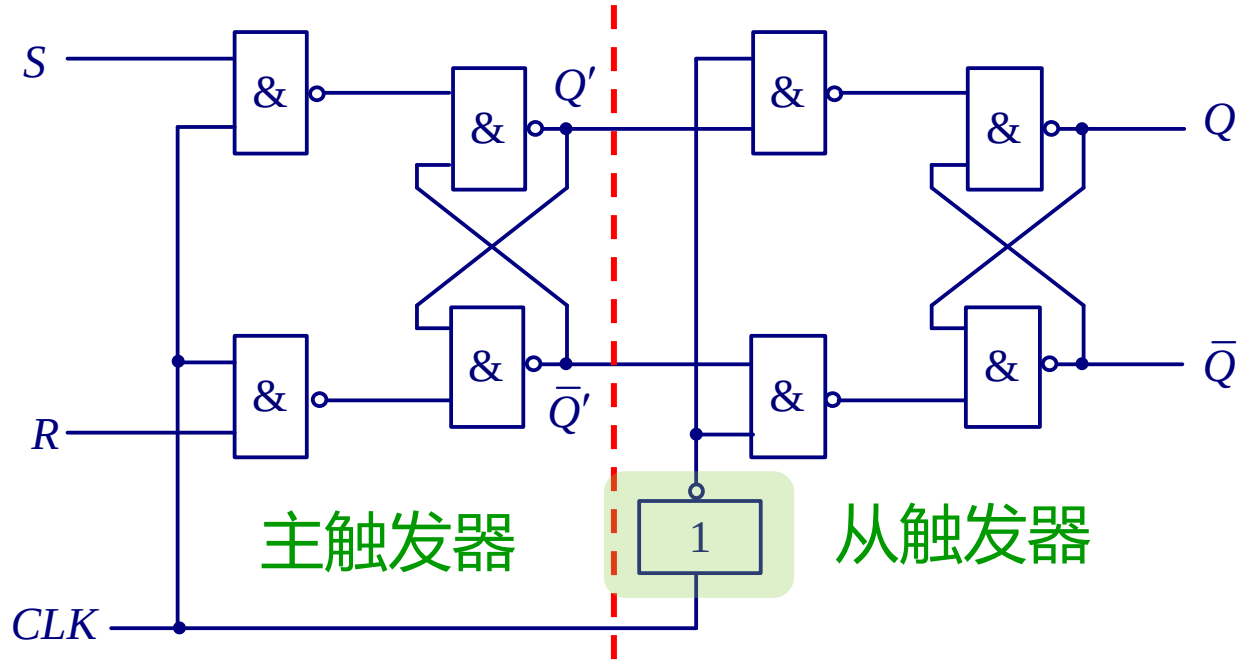
EN	D	Q^{n+1}
0	×	Q^n
1	1	1
1	0	0

例：画出门控S-R锁存器的输出波形 。假设Q的初始状态为 0。



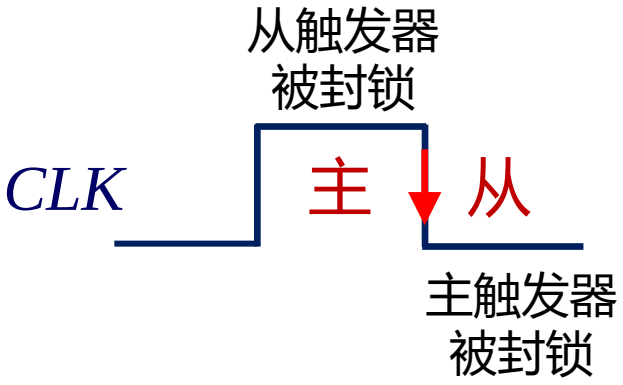
门控锁存器的输出在锁存器启动期间会随着输入的改变作相应的变化

4、主从S-R触发器 Master-Slave Flip-Flops/pulse-triggered FF



主从S-R触发器的特性表

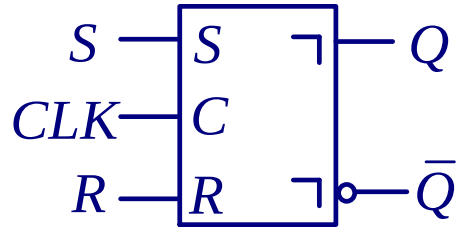
CLK	R	S	Q^{n+1}
×	0	0	Q^n
↓	0	1	1
↓	1	0	0
↓	1	1	1^*

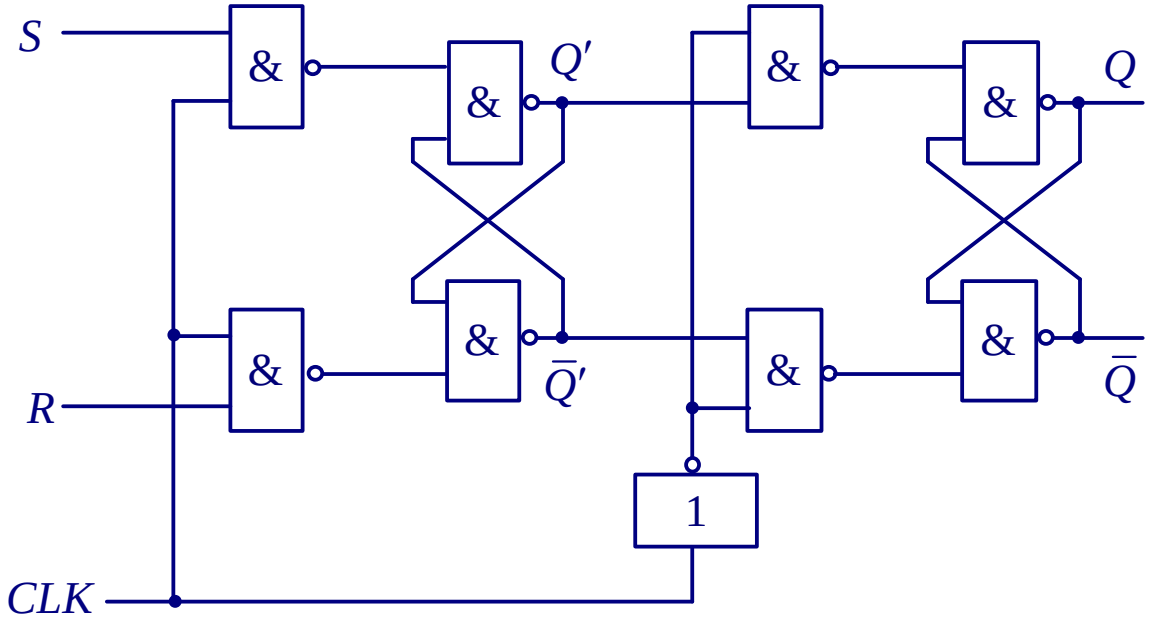


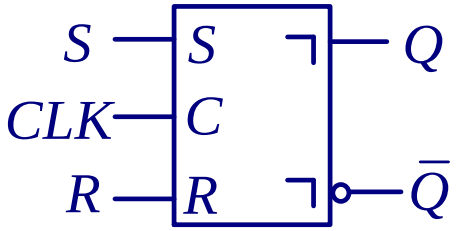
主触发器开启期间，其输出受输入的控制可能多次翻转

主从触发器输出状态的变化发生在主、从触发器交替时

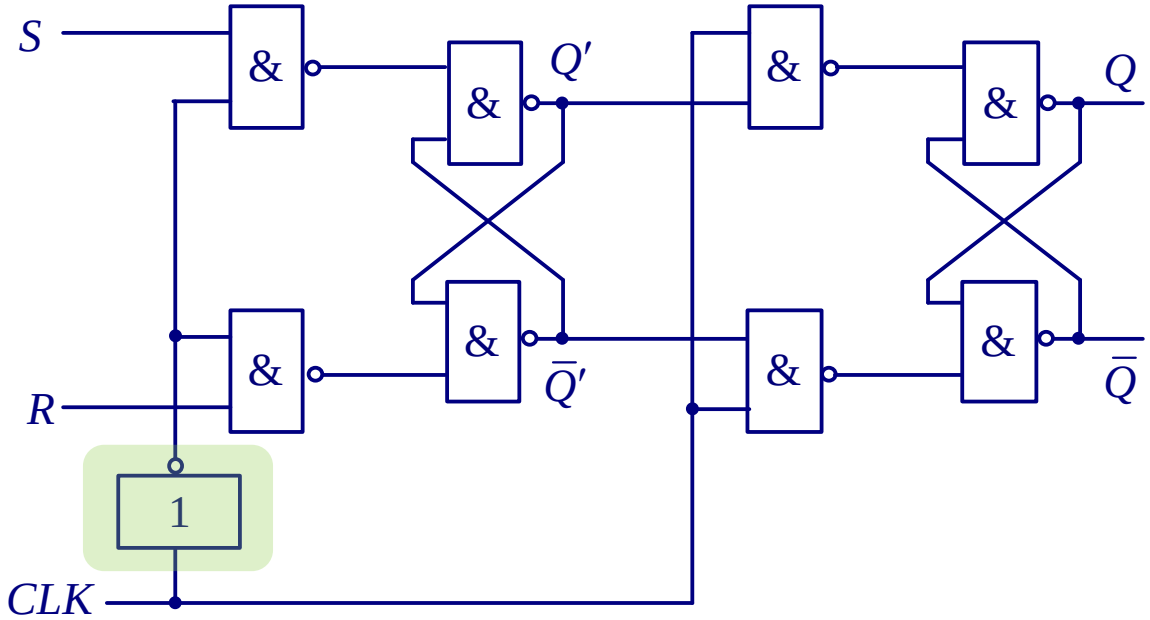
逻辑符号

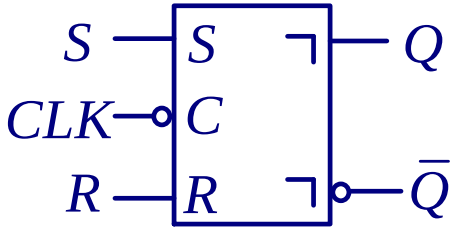






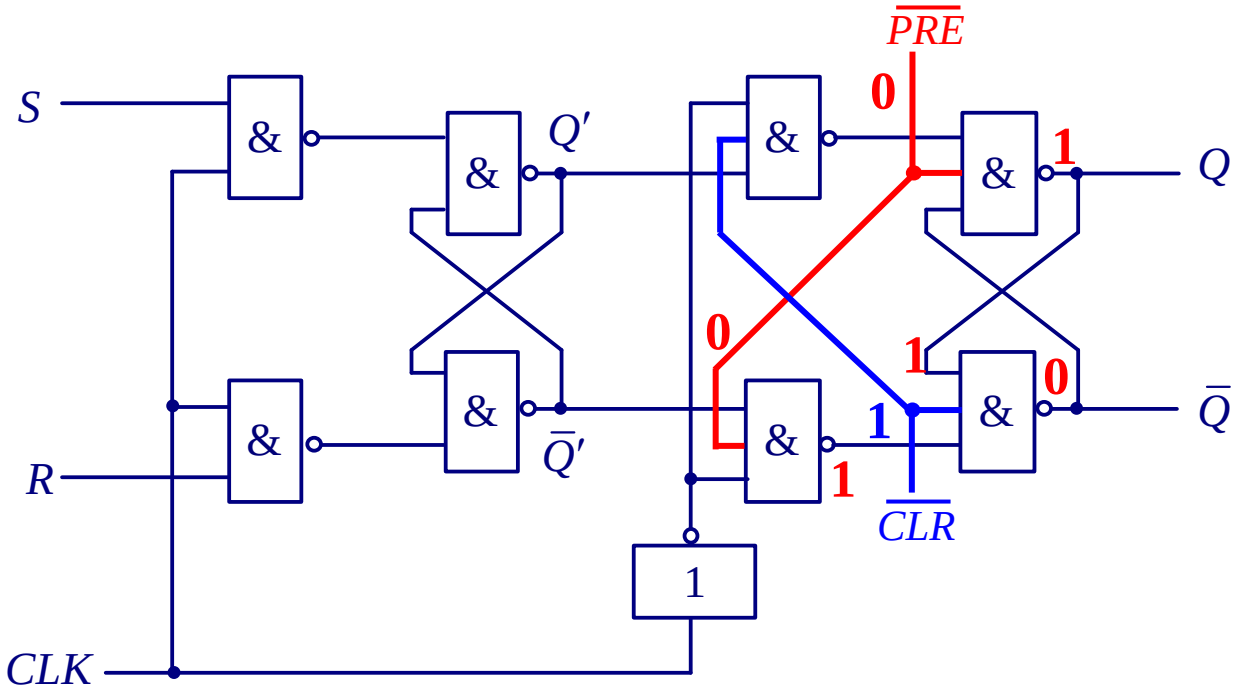
CLK	R	S	Q^{n+1}
×	0	0	Q^n
↓	0	1	1
↓	1	0	0
↓	1	1	1^*





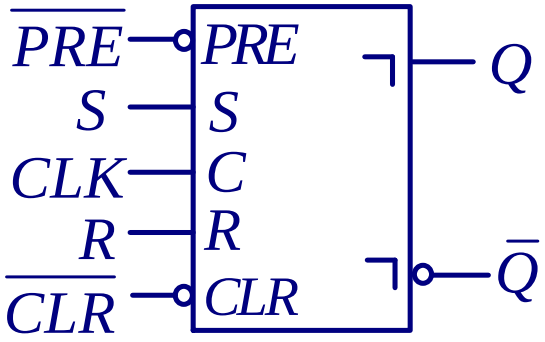
CLK	R	S	Q^{n+1}
×	0	0	Q^n
↑	0	1	1
↑	1	0	0
↑	1	1	1^*

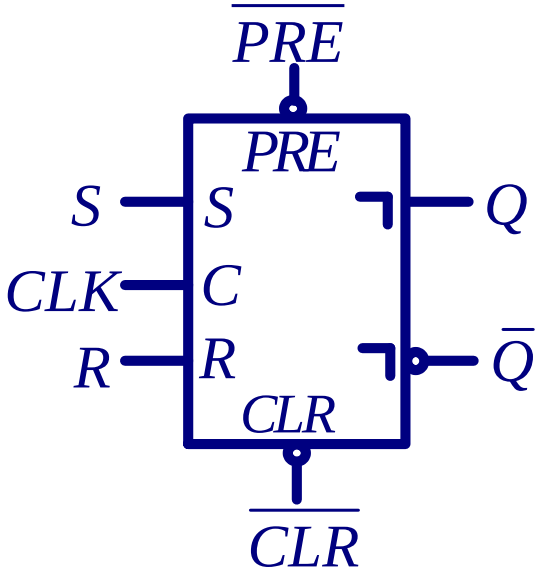
异步置位和异步复位 Asynchronous Preset and Clear inputs



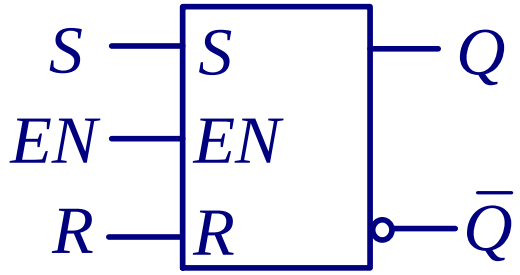
直接置位端与直接复位端不能同时有效

逻辑符号

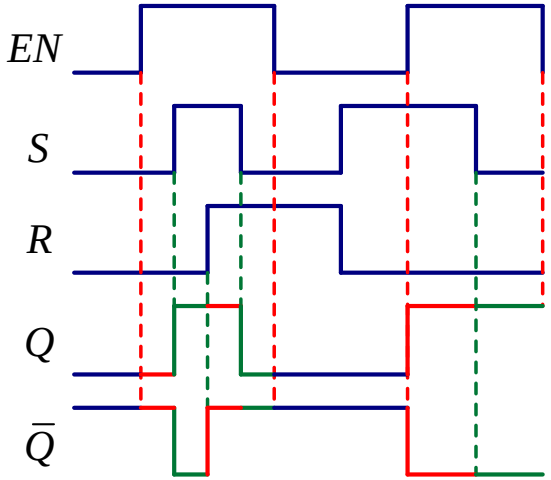




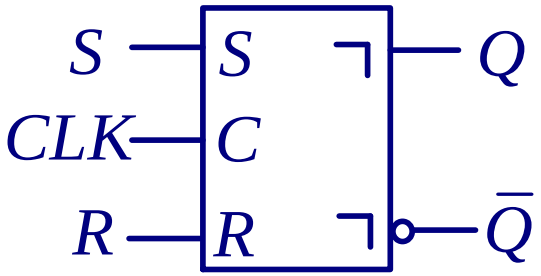
门控S-R锁存器



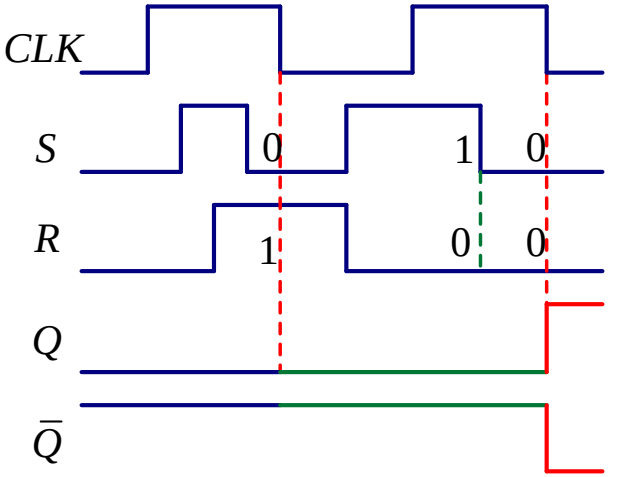
EN	R	S	Q^{n+1}
0	×	×	Q^n
1	0	0	Q^n
1	0	1	1
1	1	0	0
1	1	1	1^*



主从S-R触发器

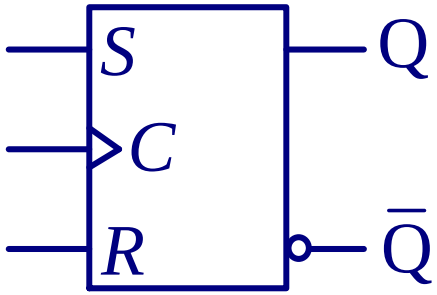


CLK	R	S	Q^{n+1}
×	0	0	Q^n
↓	0	1	1
↓	1	0	0
↓	1	1	1^*



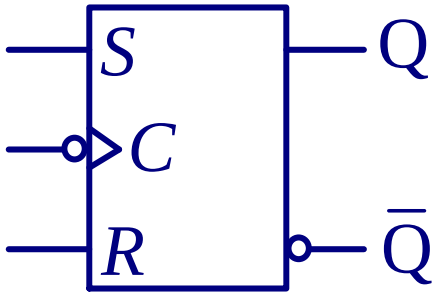
5、边沿触发器 Edge-triggered Flip-Flops

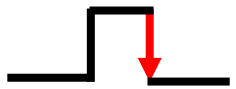
边沿触发器的次态只取决于时钟脉冲上升/下降沿到达时刻输入信号的状态





脉冲上升沿

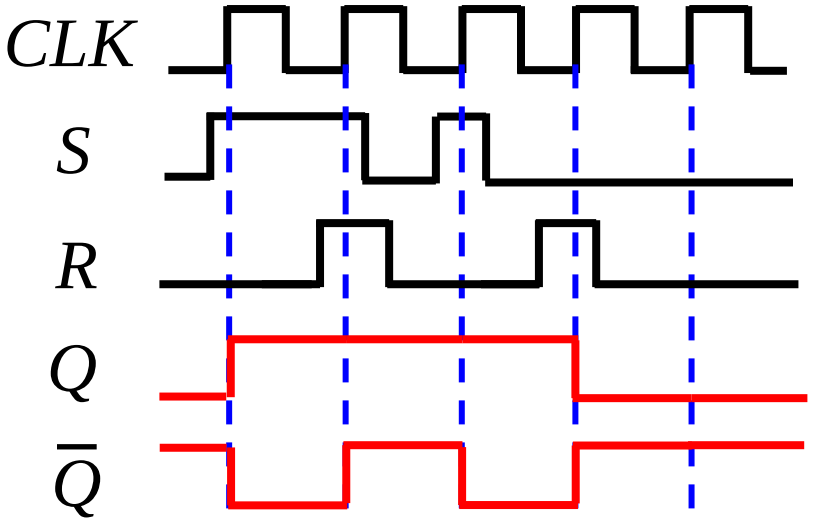




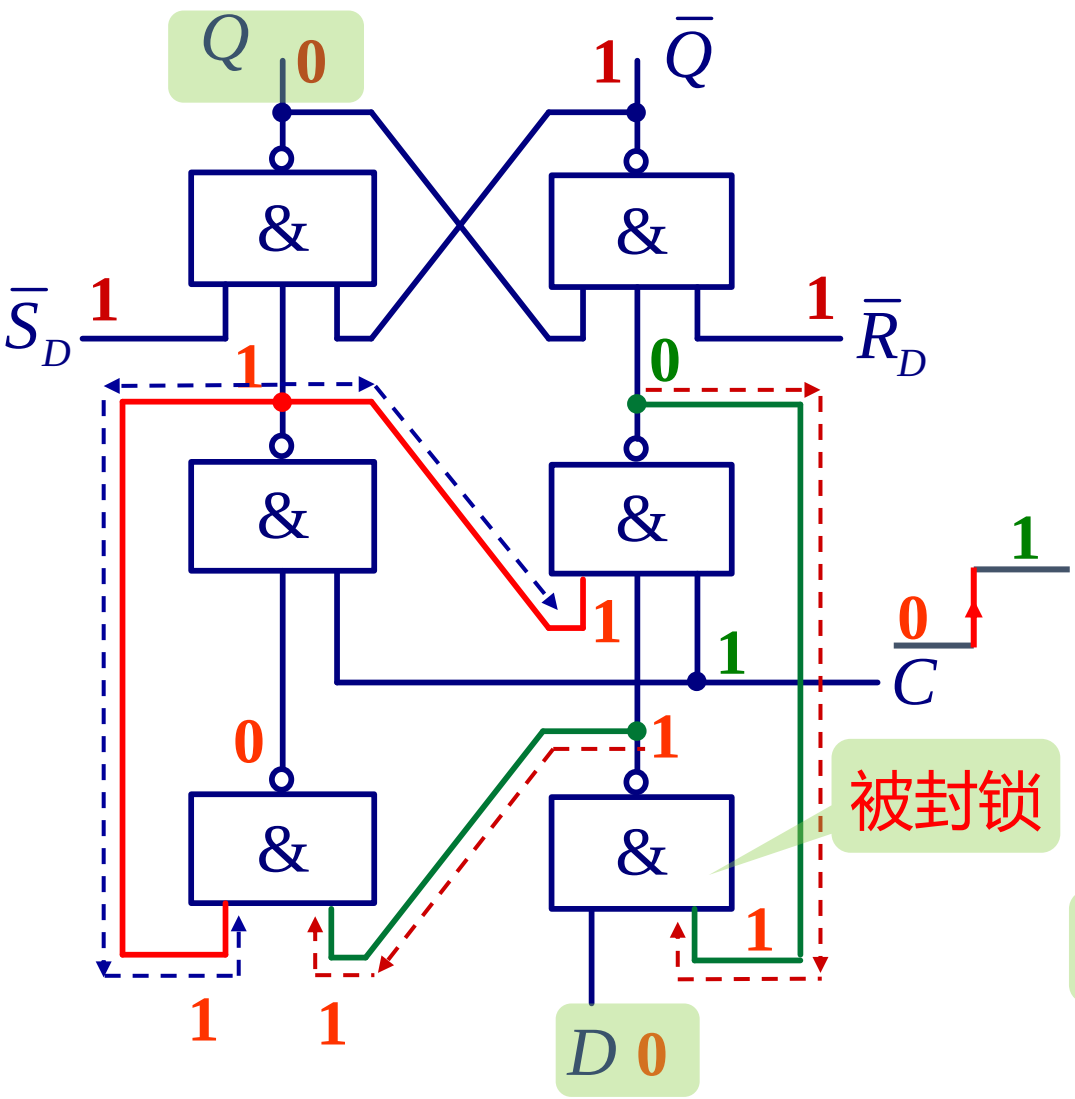
脉冲下跳沿

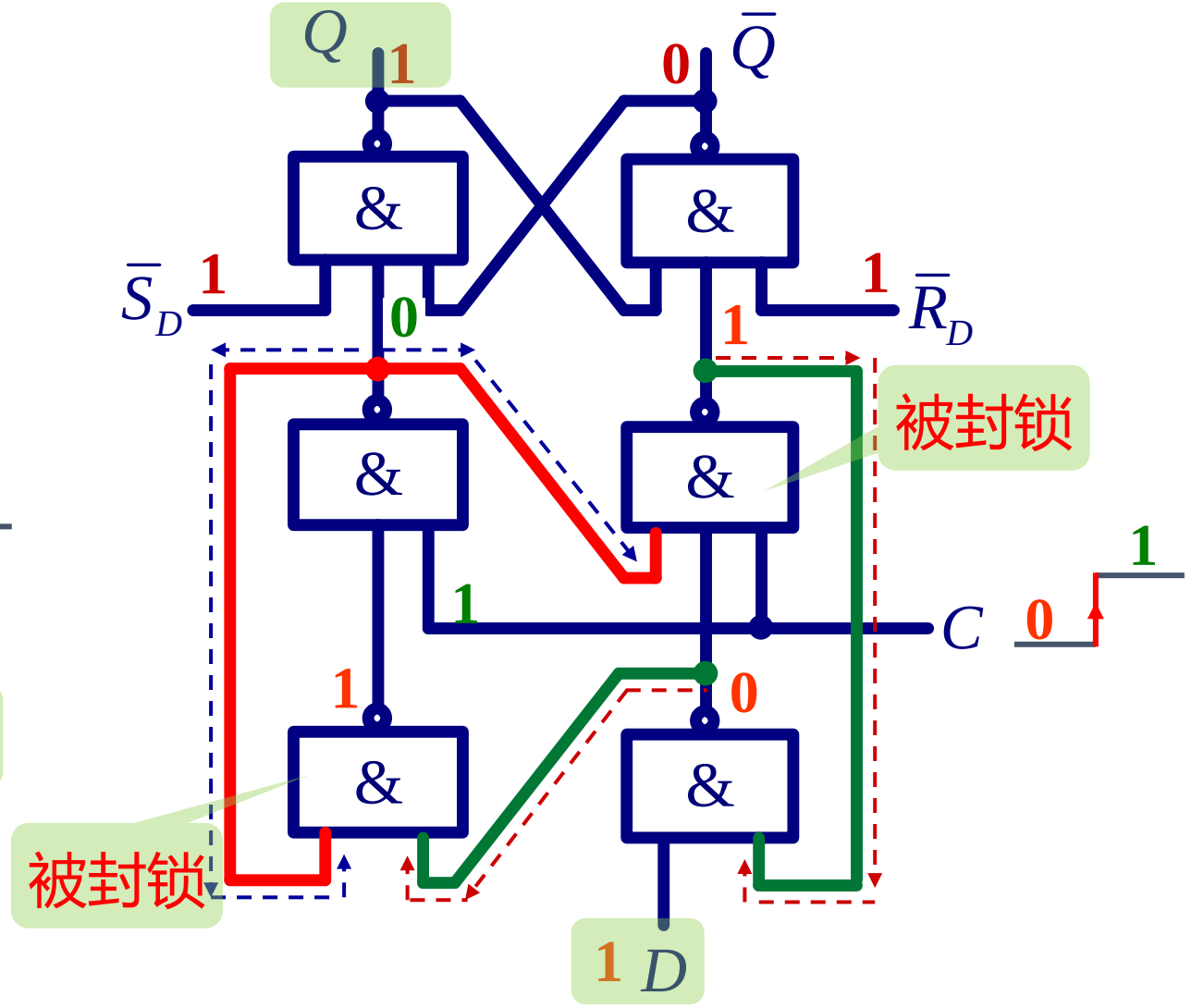
边沿RS触发器的特性表

CLK	R	S	Q^{n+1}
×	0	0	Q^n
↑	0	1	1
↑	1	0	0
↑	1	1	1^*

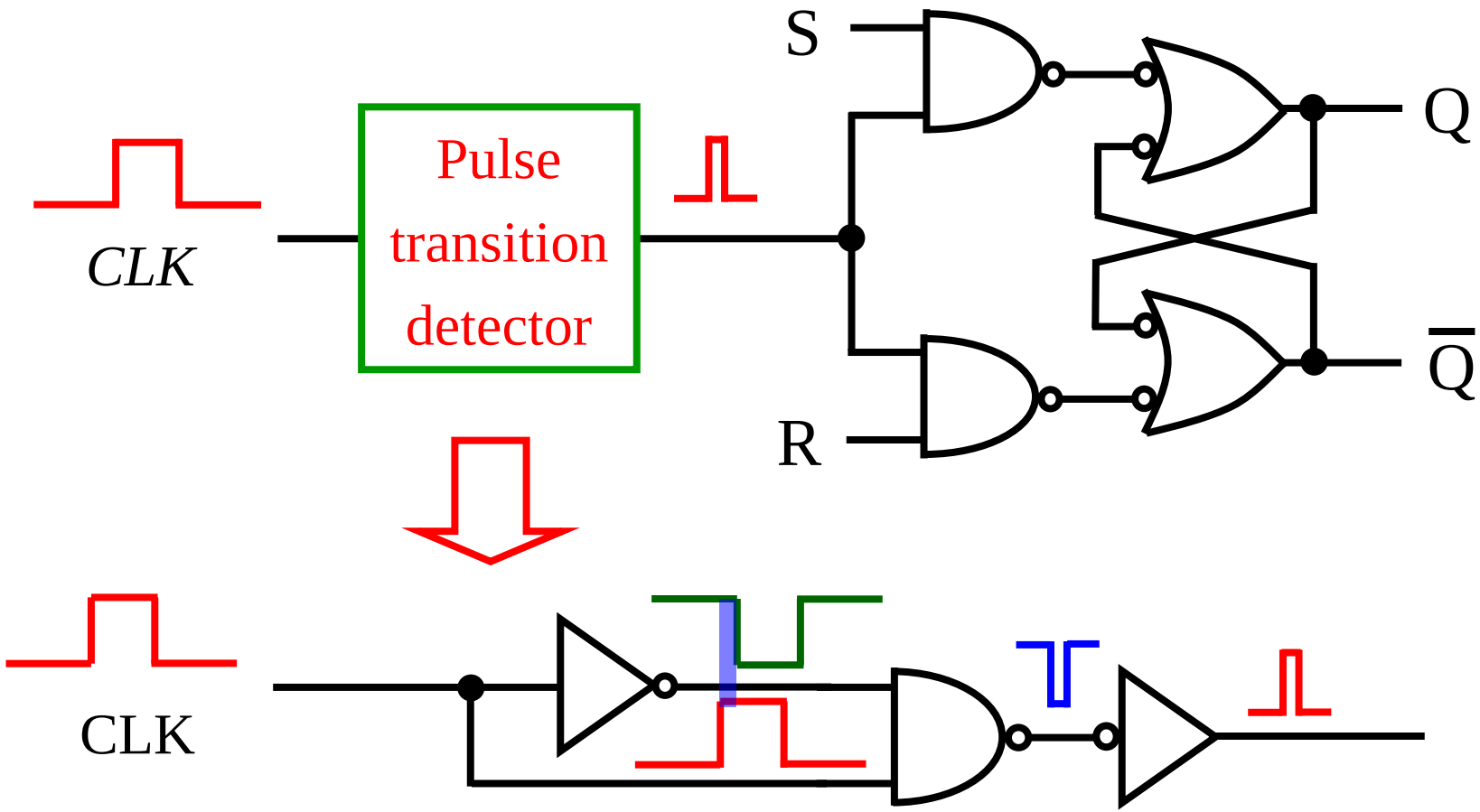


$$Q^{n+1} = D \quad (C\uparrow)$$



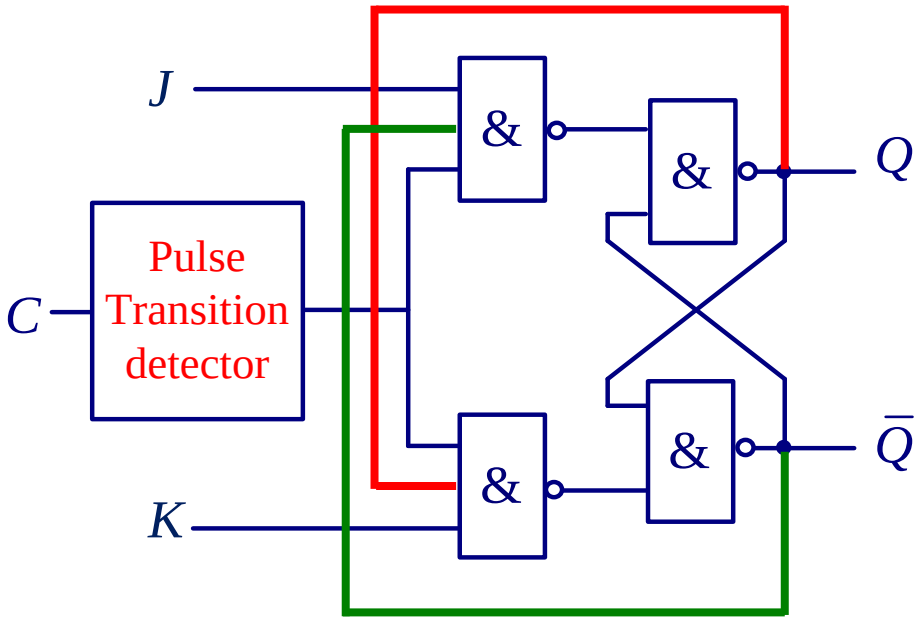


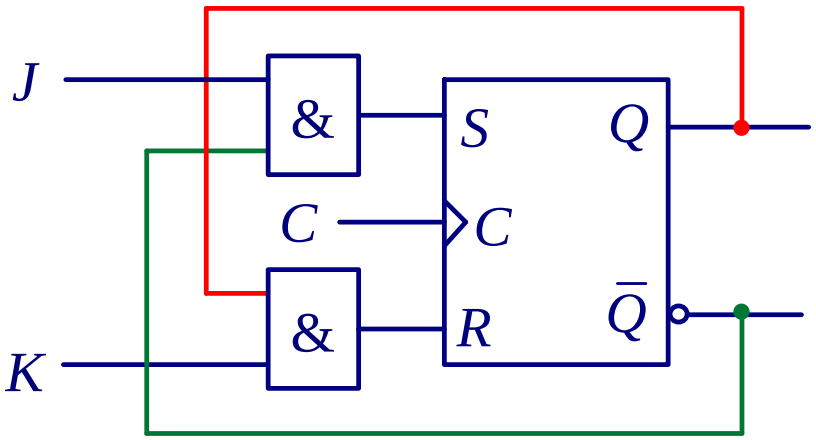
利用传输延迟的边沿触发器



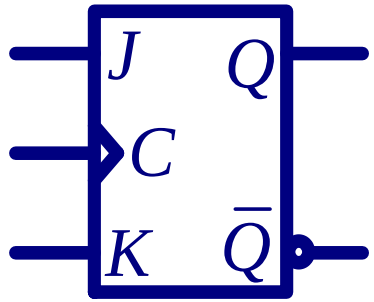
如果在脉冲的下跳沿触发，如何实现?

Edge-Triggered J-K FF





逻辑符号



$$\begin{cases} Q^{n+1} = S + \bar{R}Q^n \\ SR = 0 \end{cases}$$ 约束条件

$$S = J\bar{Q}^n$$

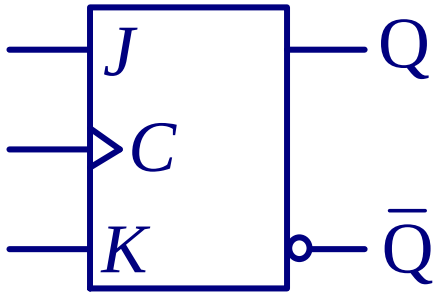
$$R = KQ^n$$

$$Q^{n+1} = S + \bar{R}Q^n = J\bar{Q}^n + (\bar{K} + \bar{Q}^n)Q^n = J\bar{Q}^n + \bar{K}Q^n$$

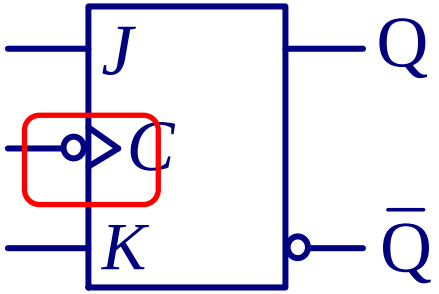
边沿JK触发器的特性表

C	J	K	Q^{n+1}
×	0	0	Q^n
↑	1	0	1
↑	0	1	0
↑	1	1	$\bar{Q}^n$

Edge-Triggered J-K FF

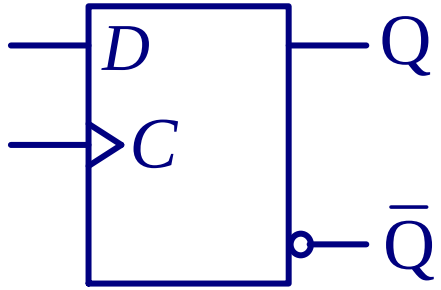


C	J	K	Q^{n+1}
×	0	0	Q^n
↑	1	0	1
↑	0	1	0
↑	1	1	$\overline{Q}^n$

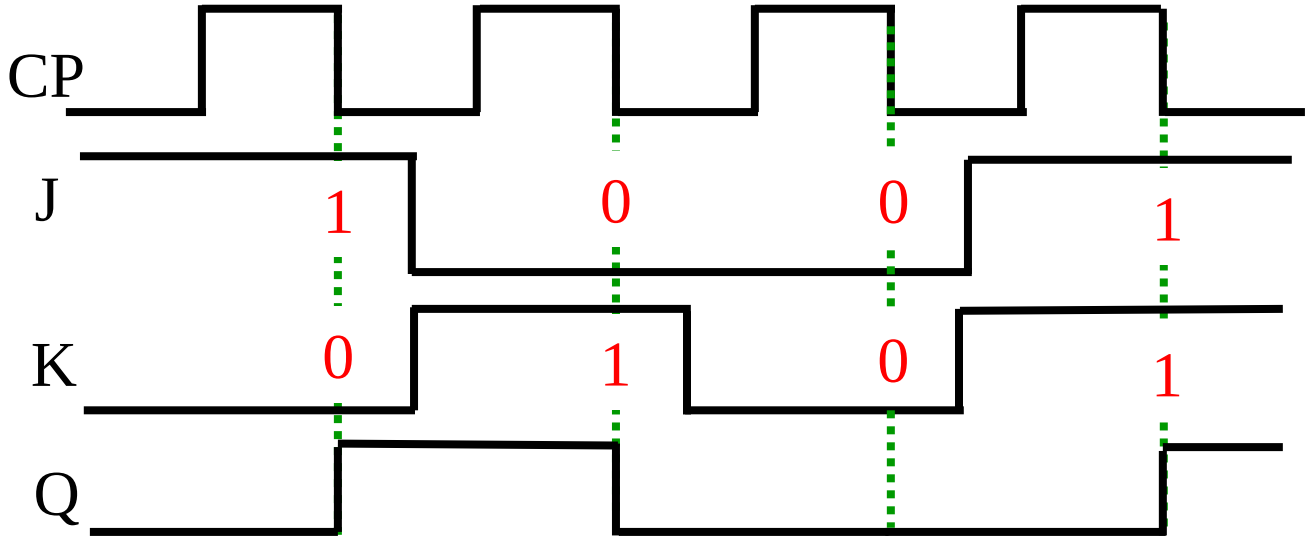


C	J	K	Q^{n+1}
×	0	0	Q^n
↓	1	0	1
↓	0	1	0
↓	1	1	$\overline{Q}^n$

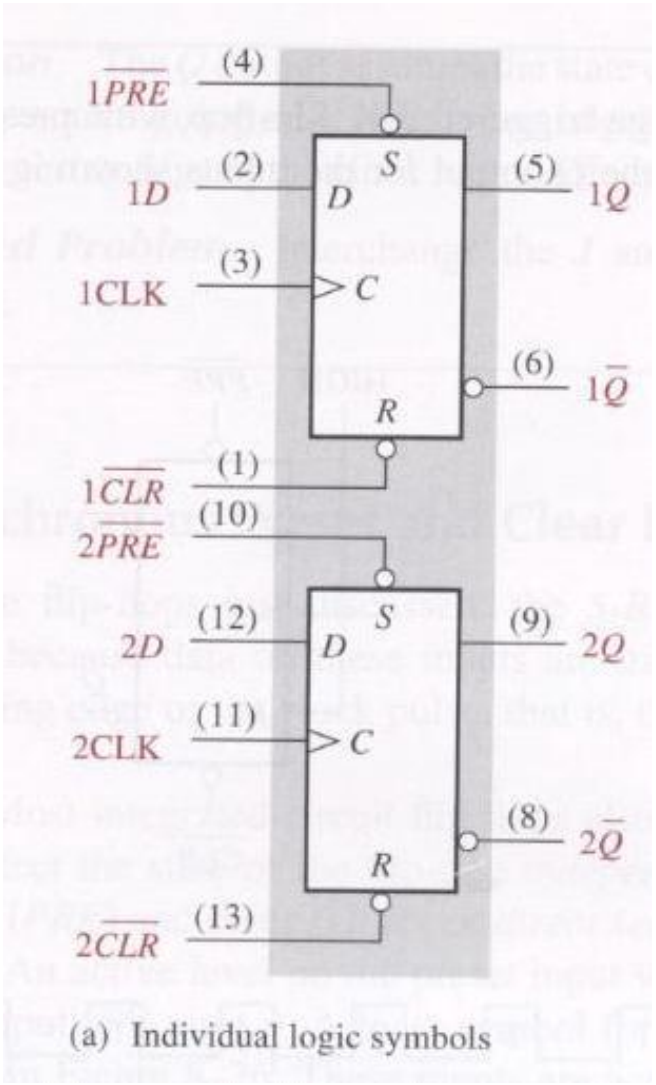
Edge-Triggered D FF



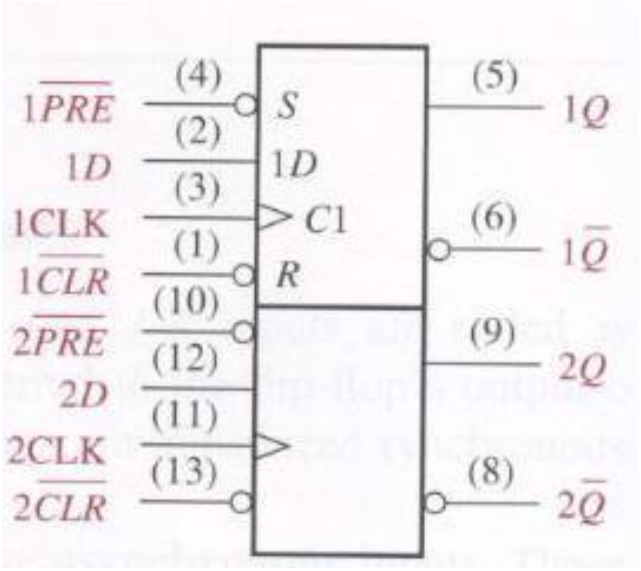
C	D	Q^{n+1}
×	×	Q^n
↑	0	0
↑	1	1



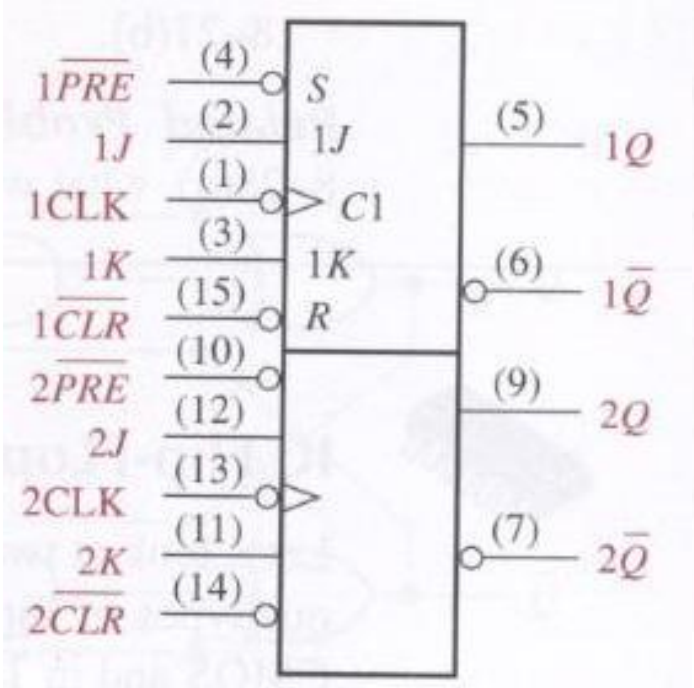
74HC74 (dual D FF)

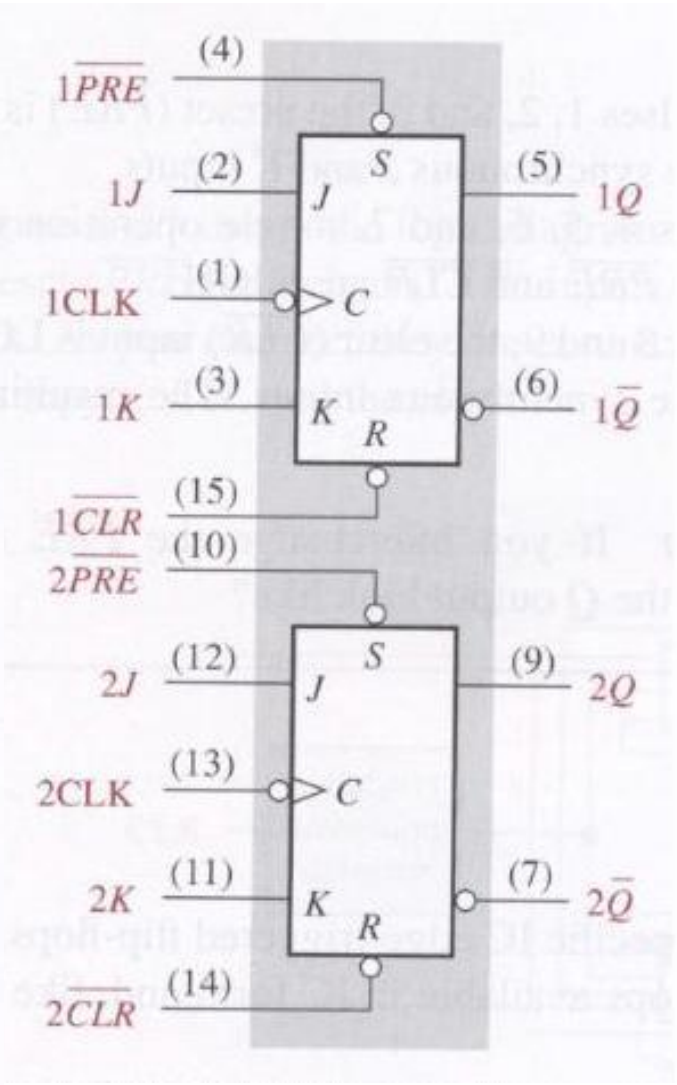


(a) Individual logic symbols

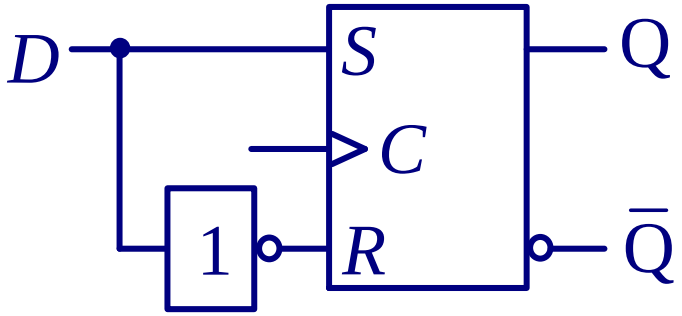


74HC112 (dual JK FF)

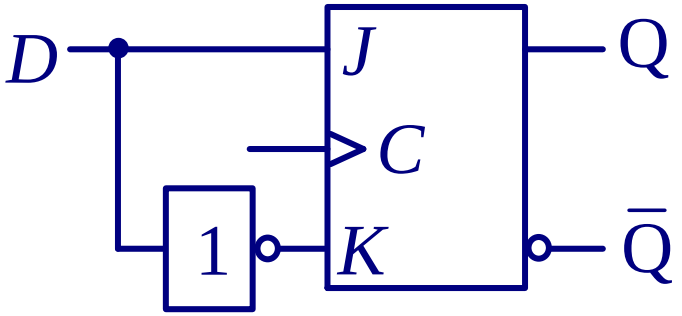




D触发器

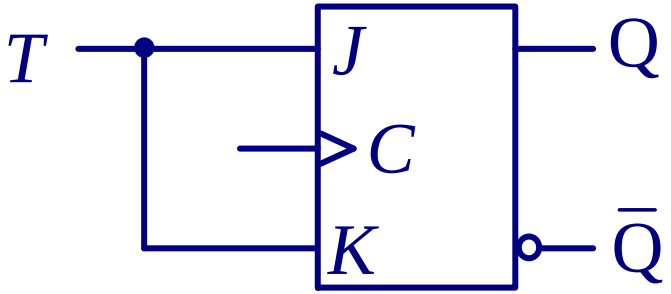


$$Q^{n+1} = D + \bar{\bar{D}}Q^n = D$$



$$Q^{n+1} = D\bar{Q}^n + \bar{\bar{D}}Q^n = D$$

T触发器



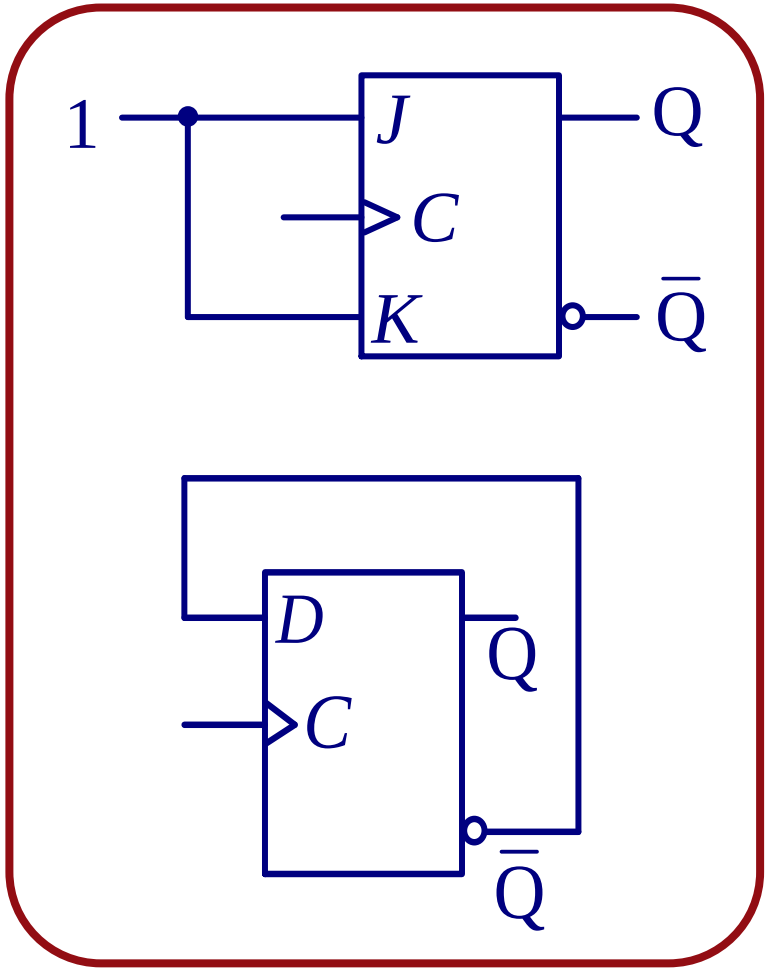
$$Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$$

$$= T\bar{Q}^n + \bar{T}Q^n$$

$T=1$ 时 $Q^{n+1} = \bar{Q}^n$

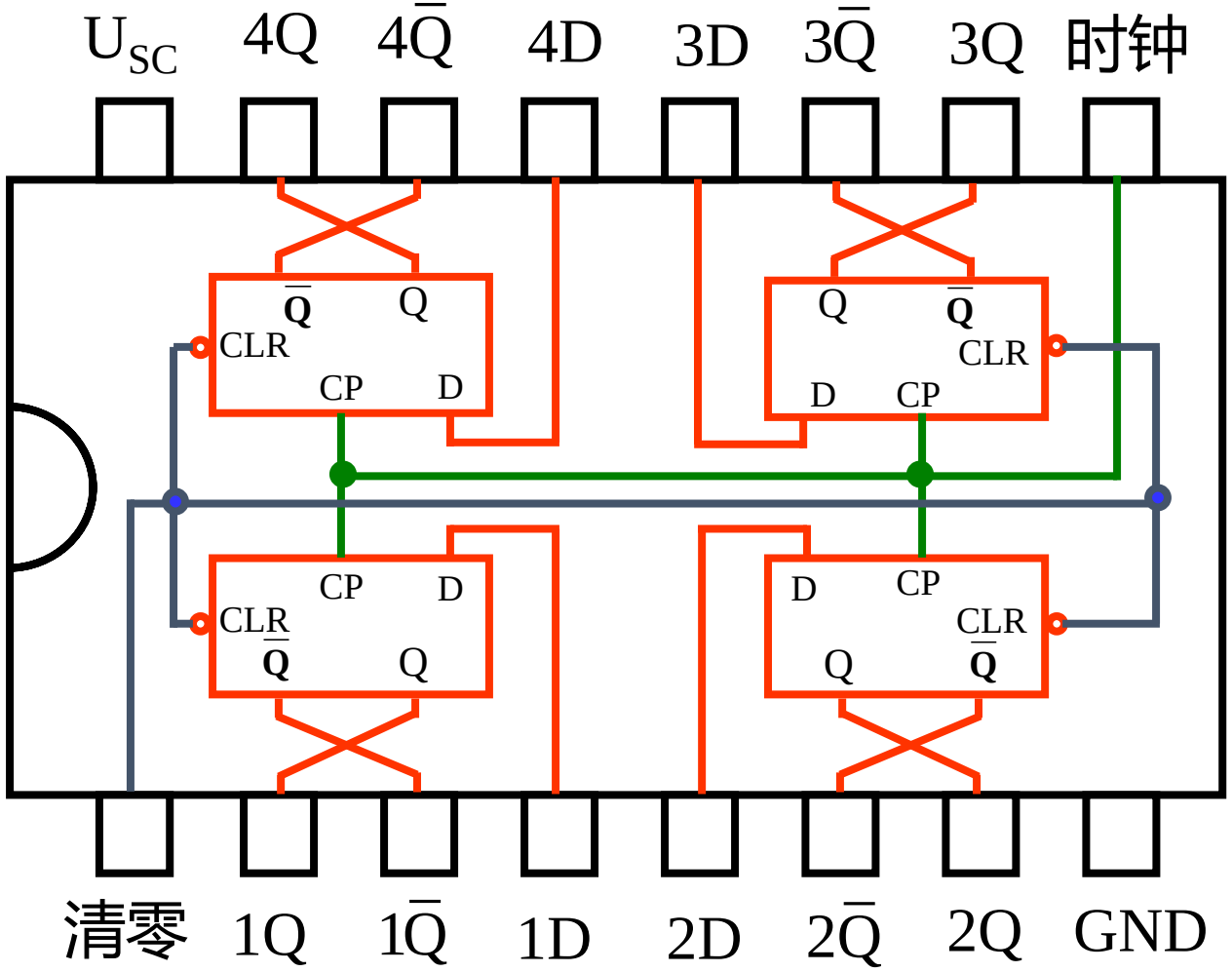
$T=0$ 时 $Q^{n+1} = Q^n$

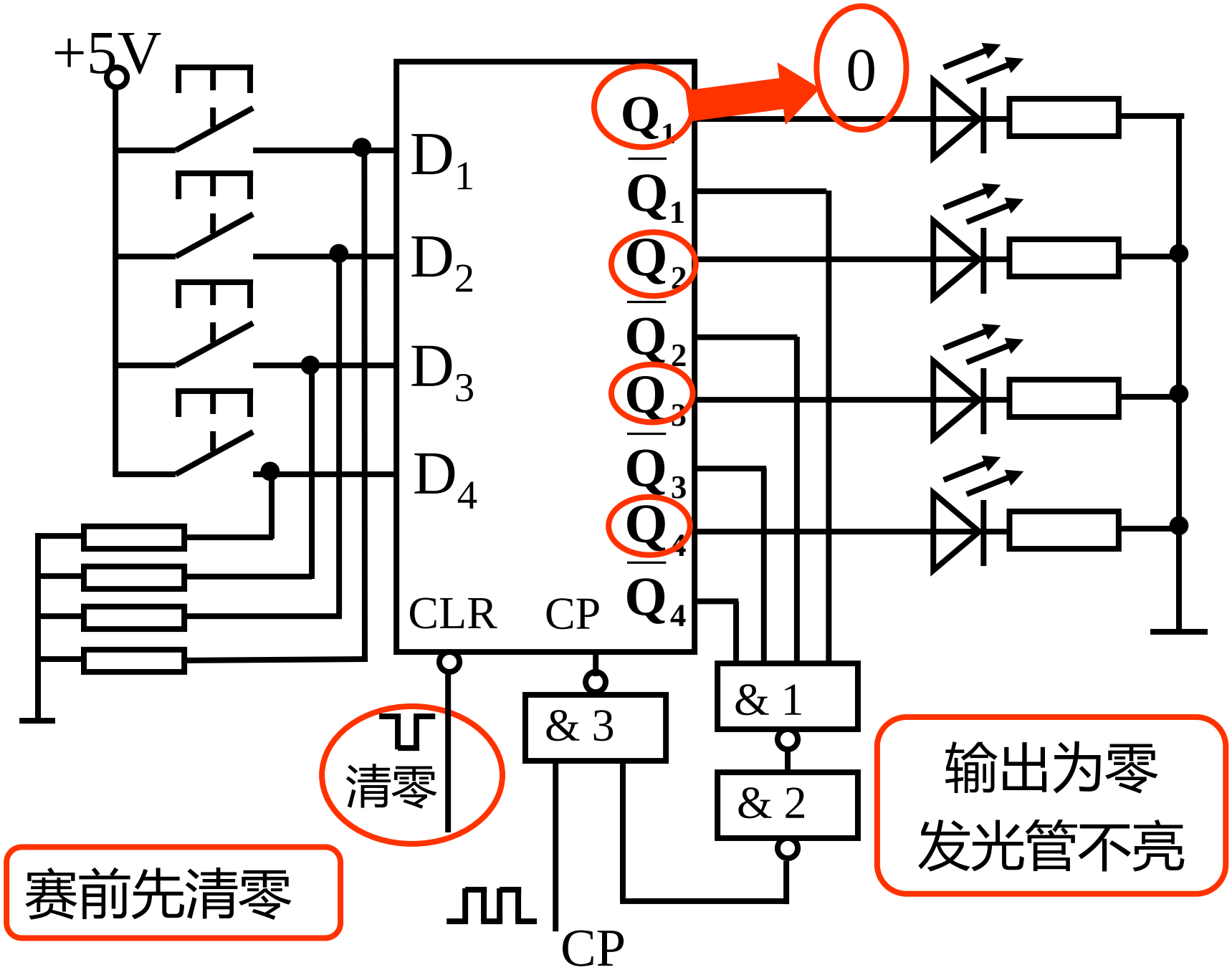
T′触发器 $Q^{n+1} = \bar{Q}^n$

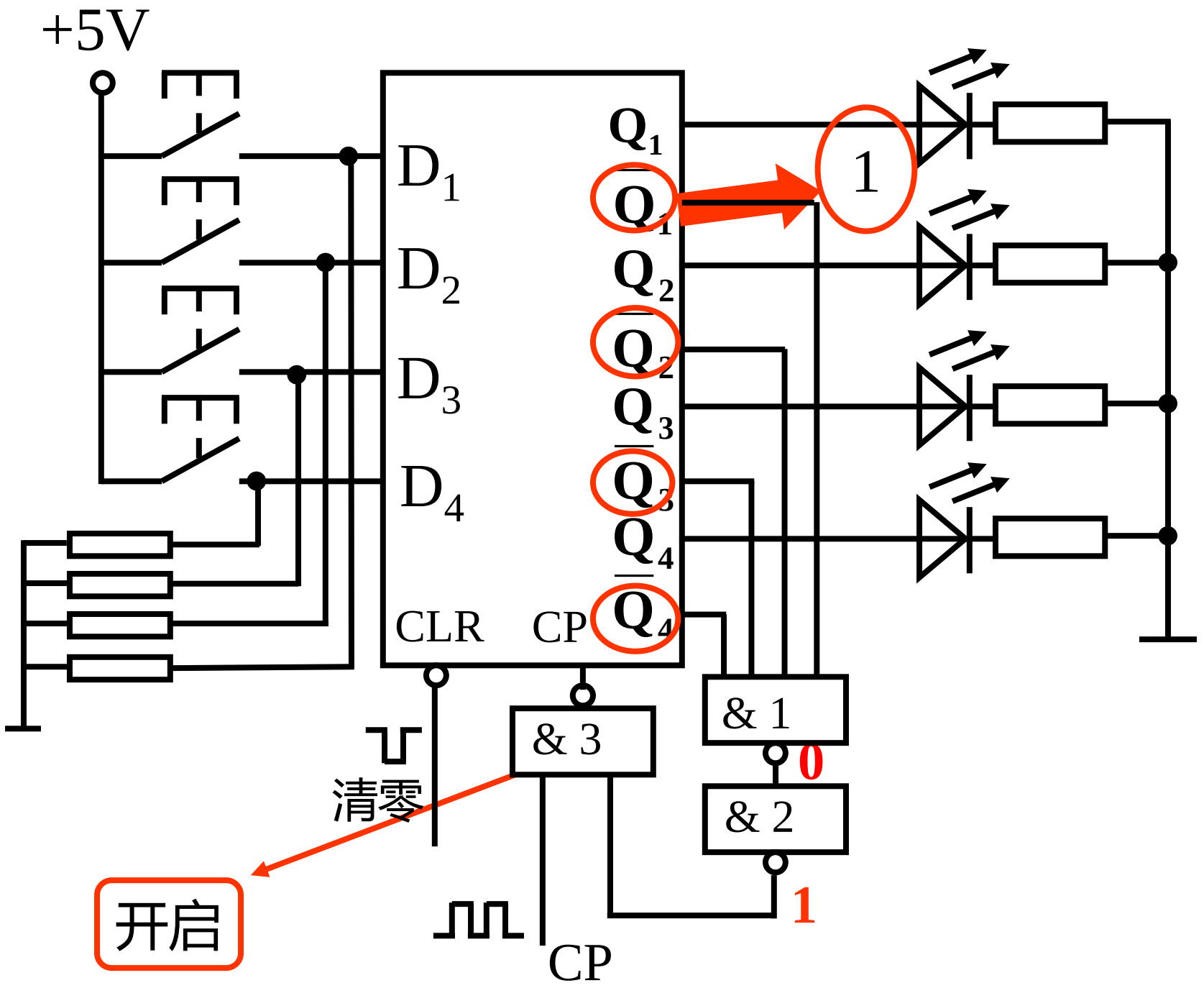


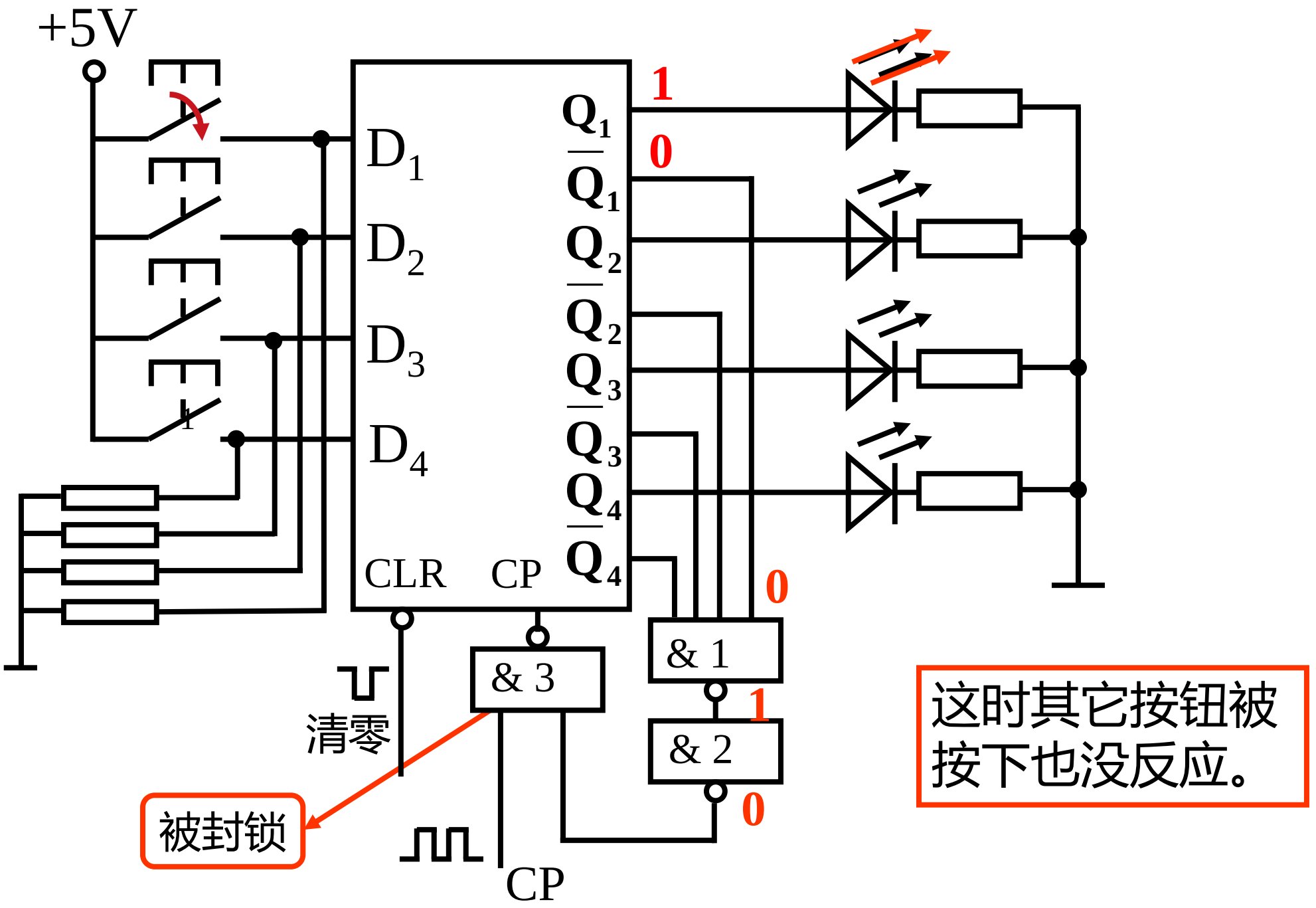
例：四人抢答电路。四人参加比赛，每人一个按钮，其中一人按下按钮后，相应的指示灯亮，且此时其它按钮按下不起作用。

电路的核心是**74LS175**四D触发器。它的内部包含了四个D触发器，各输入、输出以字头相区别，管脚图见右图。

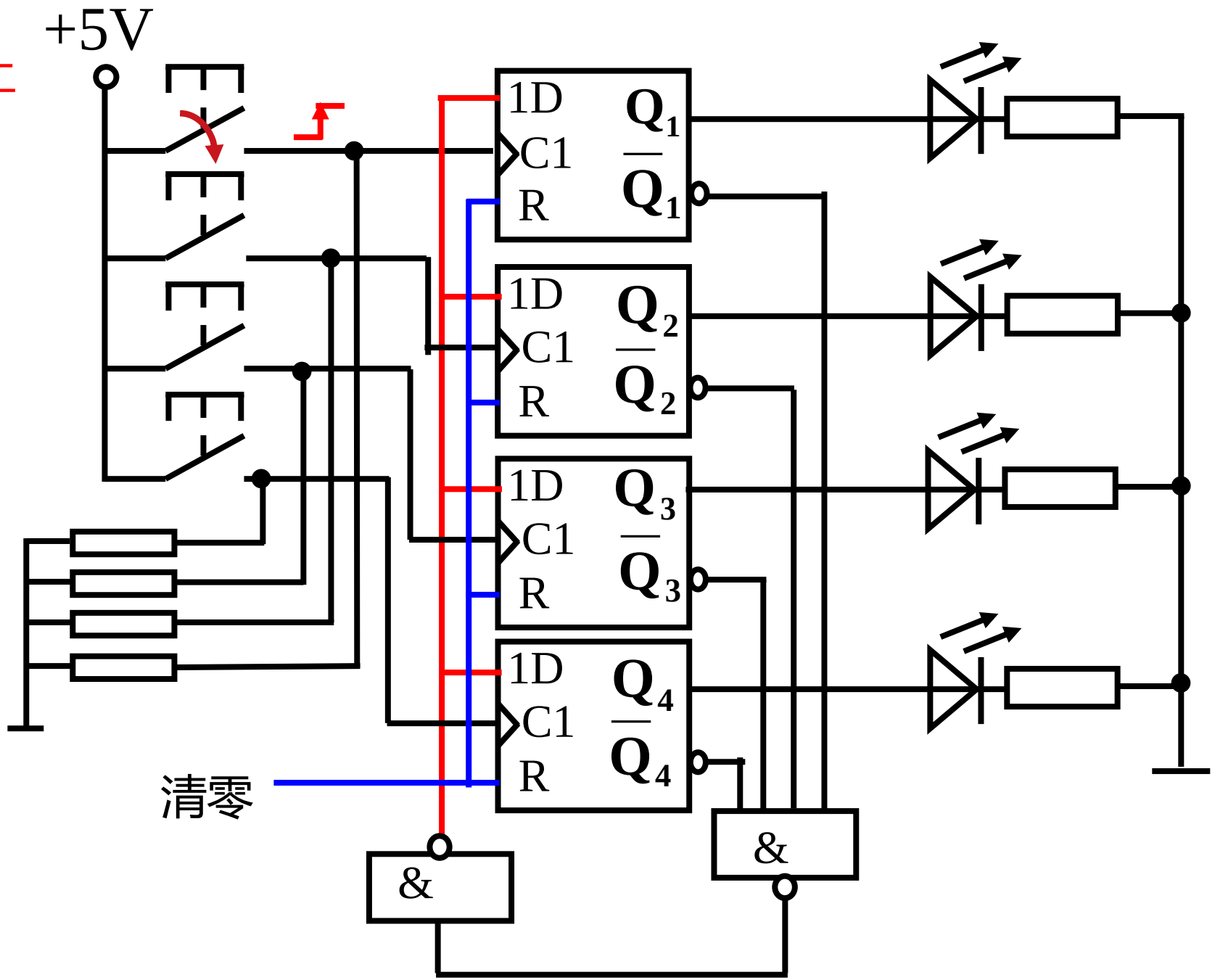



+5V
D1
D2
D3
D4
Q1
Q1
Q2
Q2
Q3
Q3
Q4
Q4
0
CLR
CP
清零
& 3
& 1
& 2
CP
赛前先清零
输出为零
发光管不亮


+5V
D1
D2
D3
D4
Q1
Q1
Q2
Q2
Q3
Q3
Q4
Q4
1
CLR
CP
清零
& 3
& 1
0
& 2
1
开启
CP


+5V
E
E
E
E
1
D1
D2
D3
D4
Q1
Q1
Q2
Q2
Q3
Q3
Q4
Q4
1
0
CLR
CP
0
& 1
1
& 2
0
& 3
清零
被封锁
CP
这时其它按钮被
按下也没反应。

解法二


+5V
1D
C1
R
Q_1
$\overline{Q}_1$
Q_2
$\overline{Q}_2$
Q_3
$\overline{Q}_3$
Q_4
$\overline{Q}_4$
清零
&
&

6、触发器工作特性参数 Operating Characteristics

Propagation Delay Times

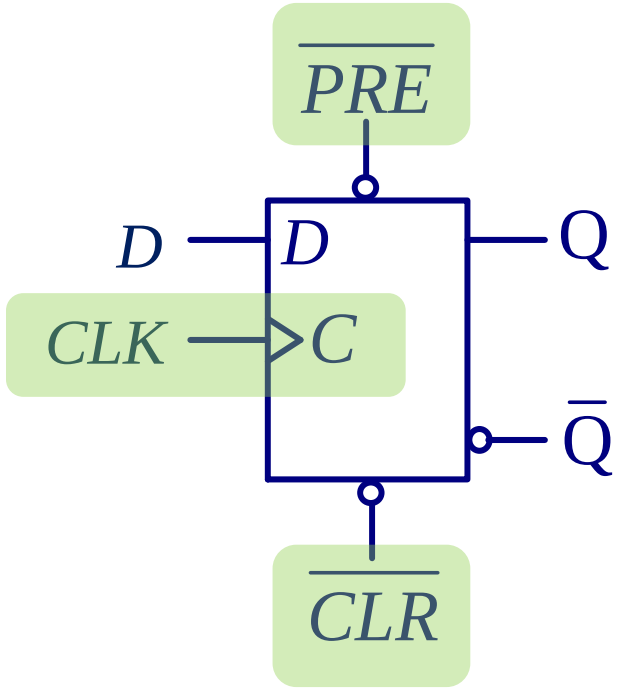
Set-up Time

Hold Time

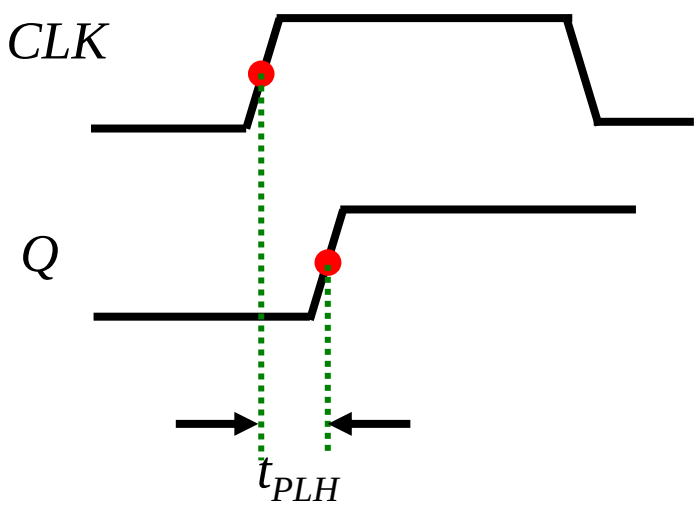
Maximum Clock Frequency

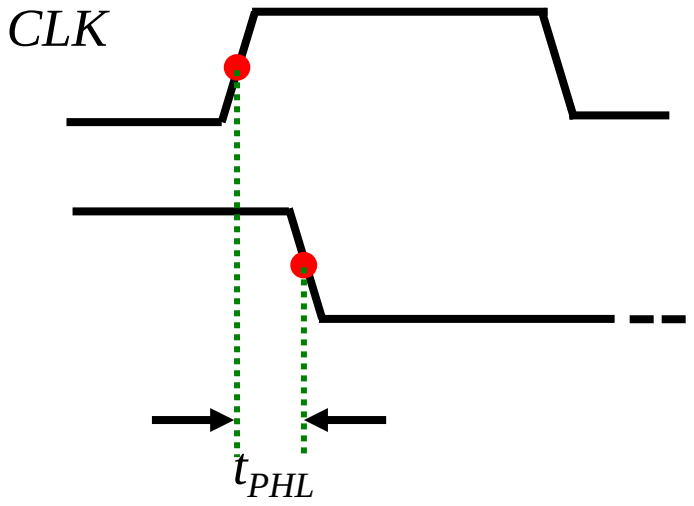
Pulse Widths

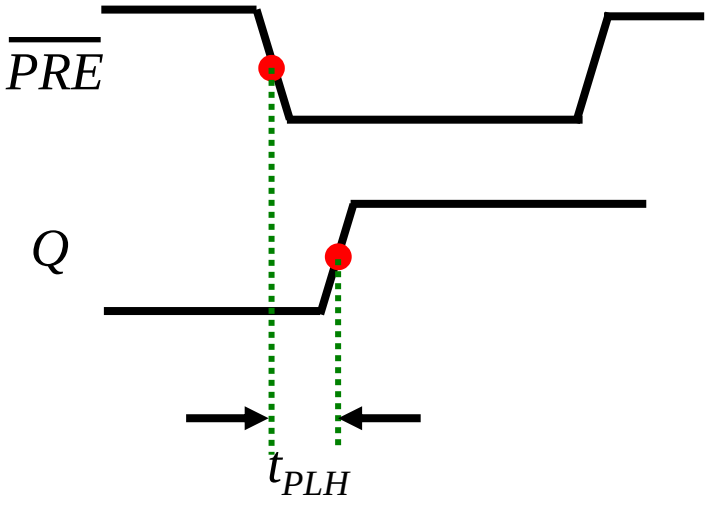
Power Dissipation

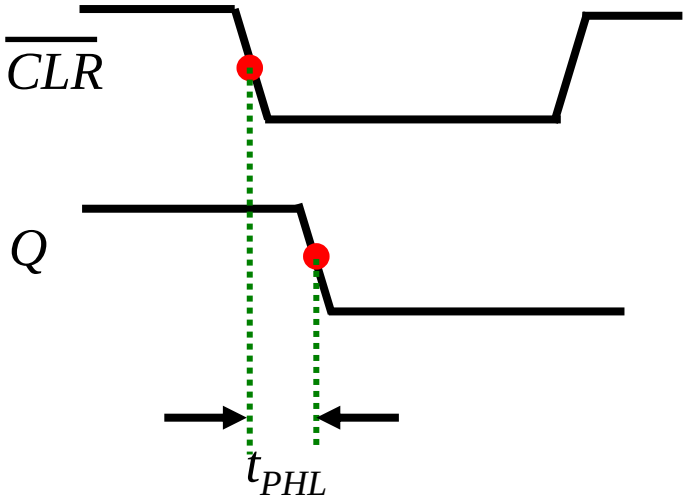


Propagation Delay Times

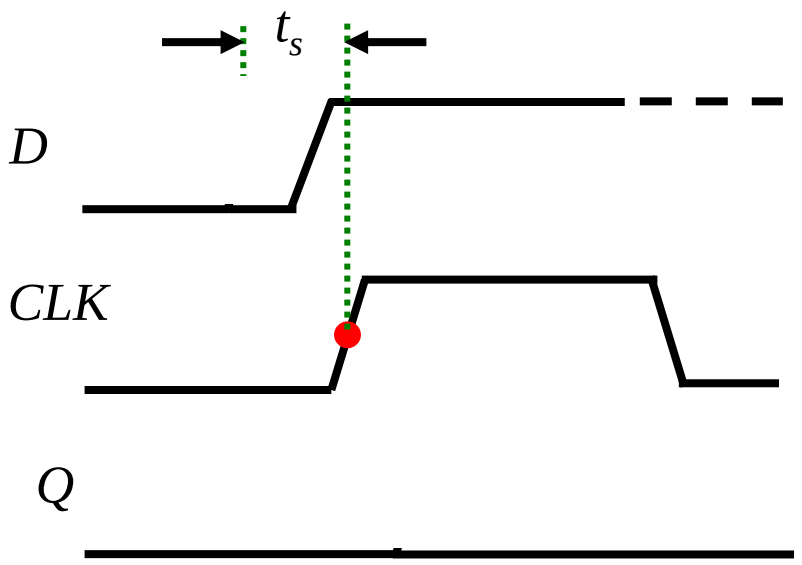




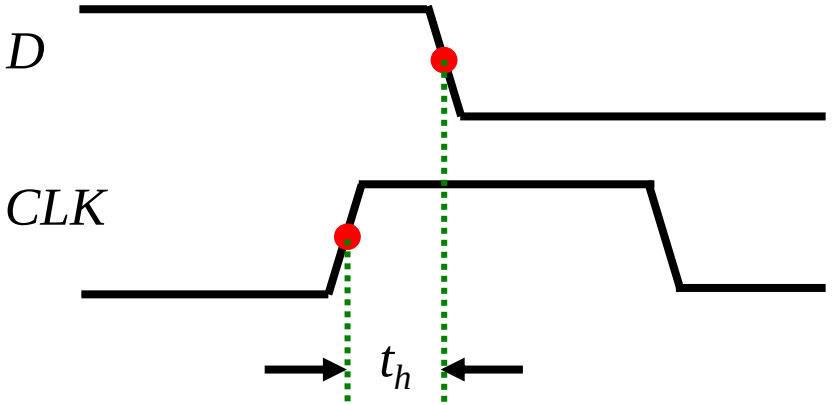




Set-up Time

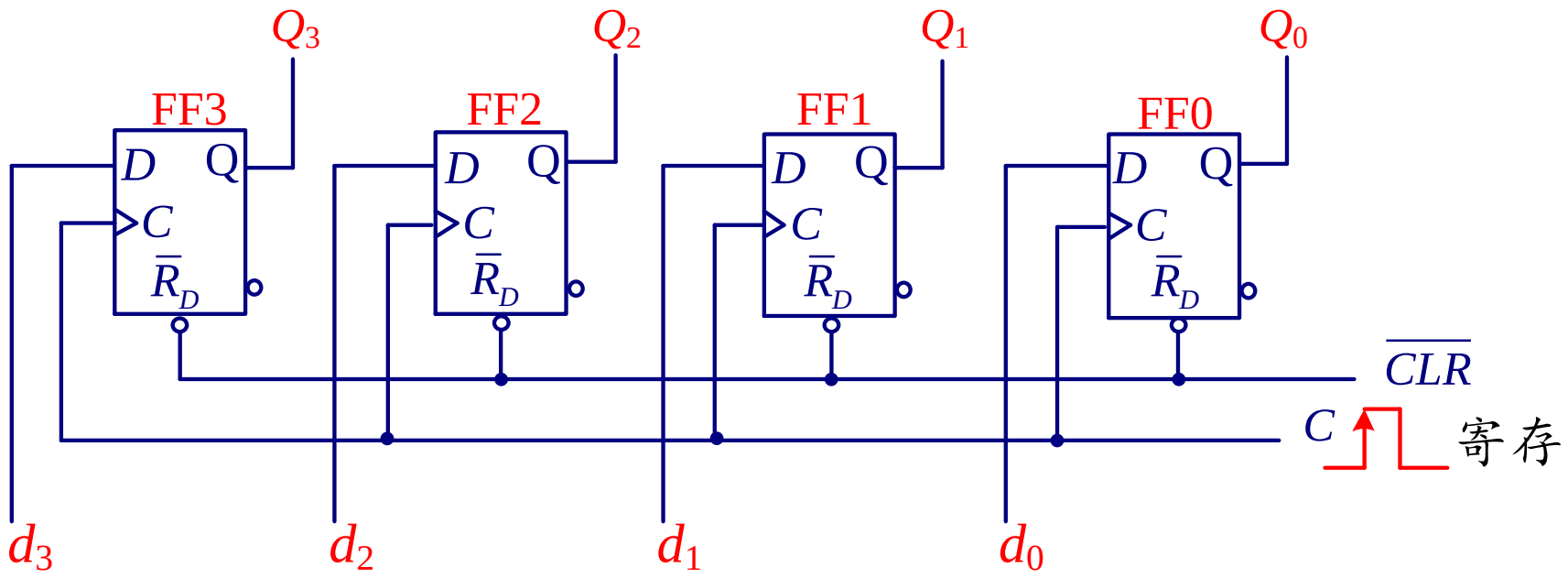


Hold Time

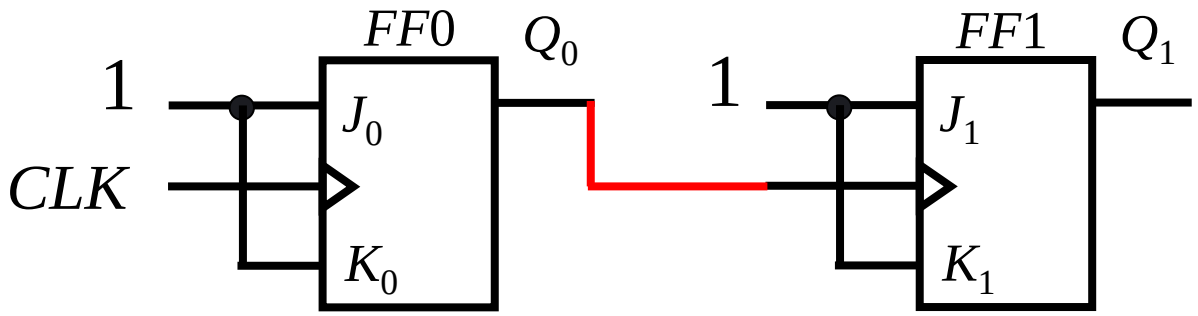


7、数码寄存器

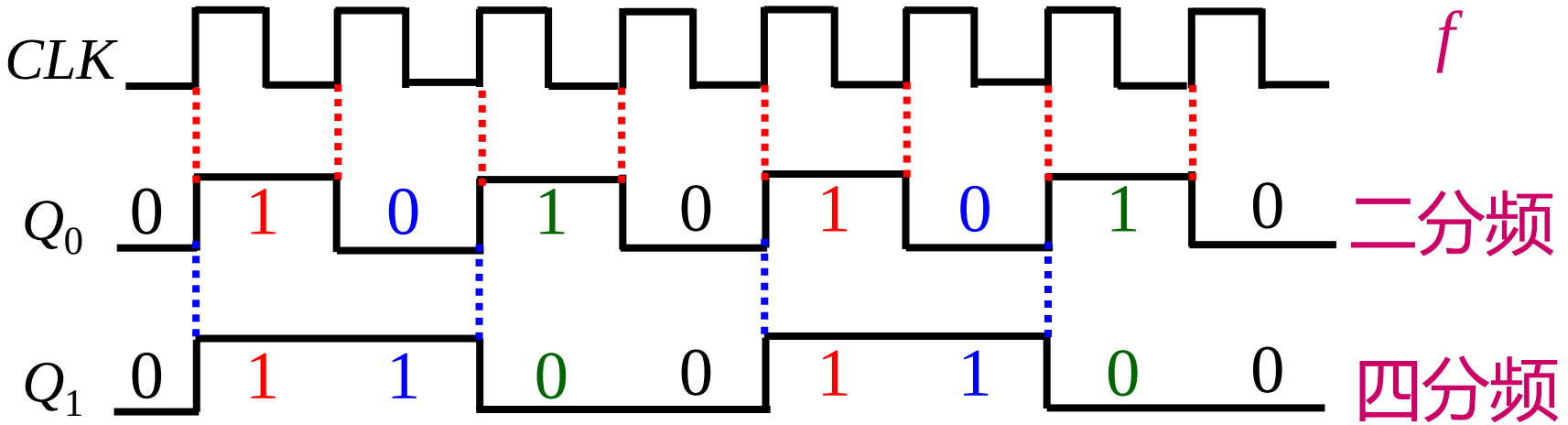
数据存储、分频、计数



四位数码寄存器

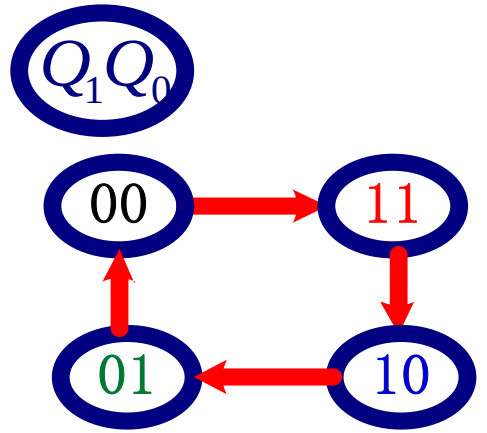


$$Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n = \bar{Q}^n$$



*n*个触发器可以实现？分频

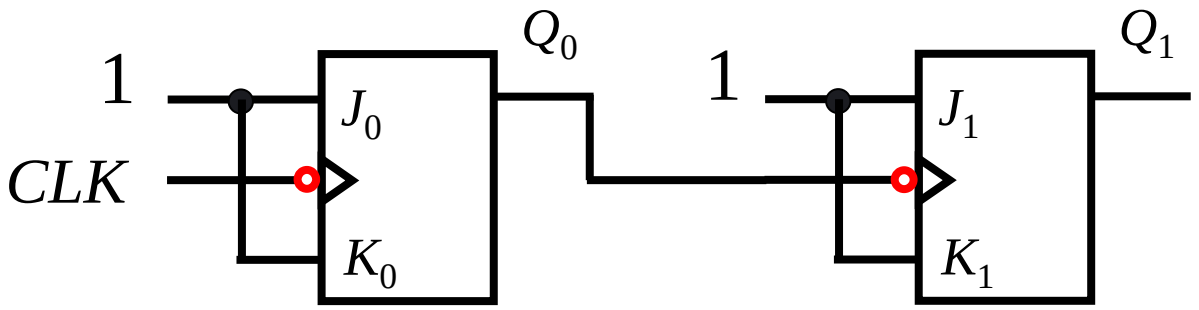
State diagram



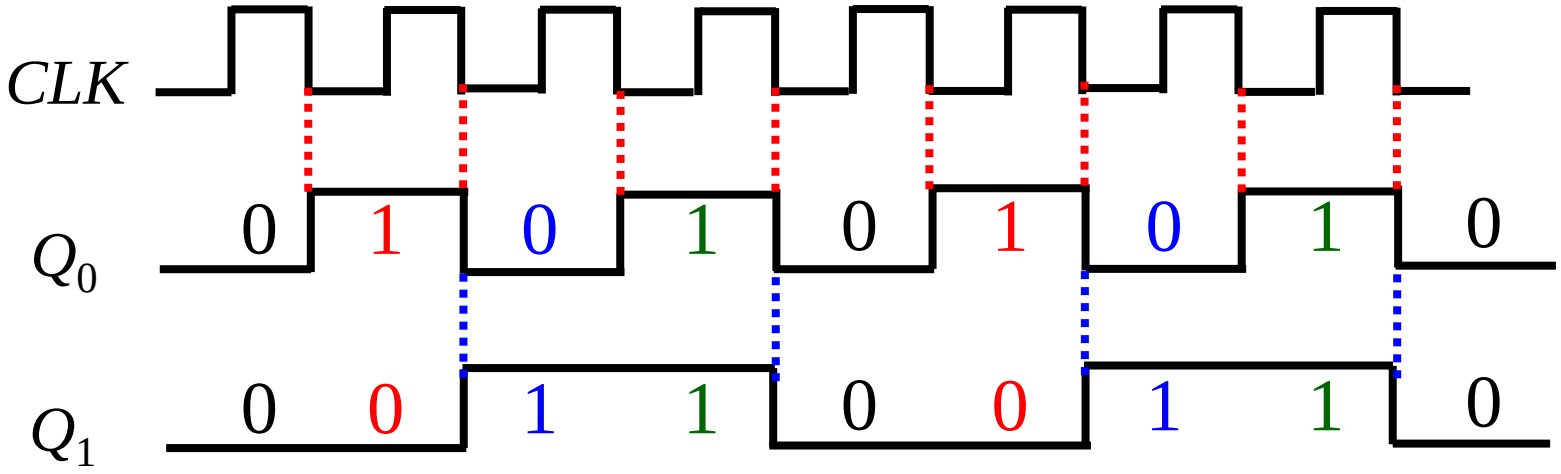
n 位二进制 2^n

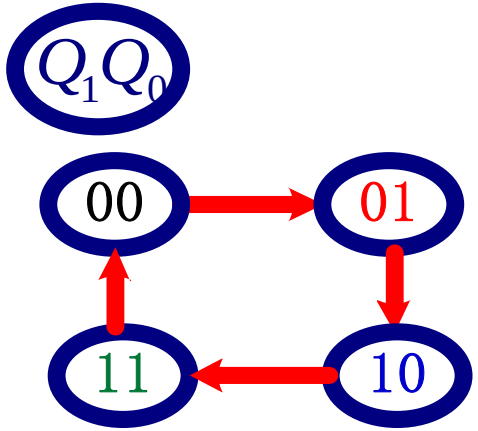
减法计数器

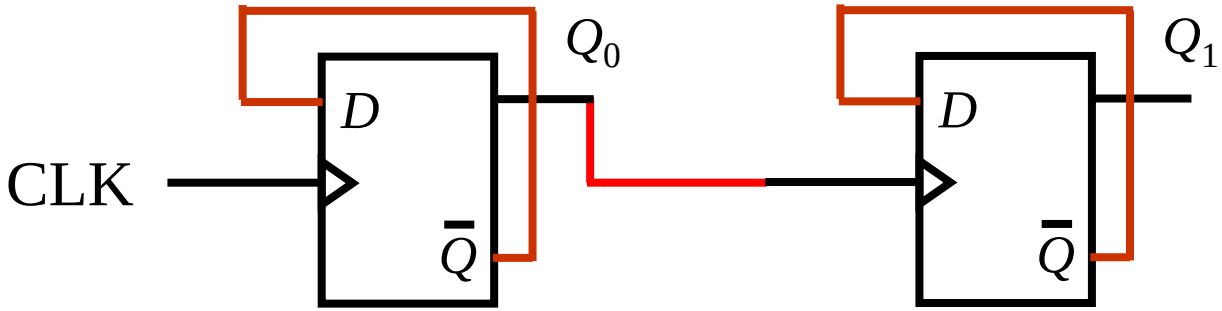
Modulo 4 Counter


1
CLK
J_0
K_0
Q_0
1
J_1
K_1
Q_1

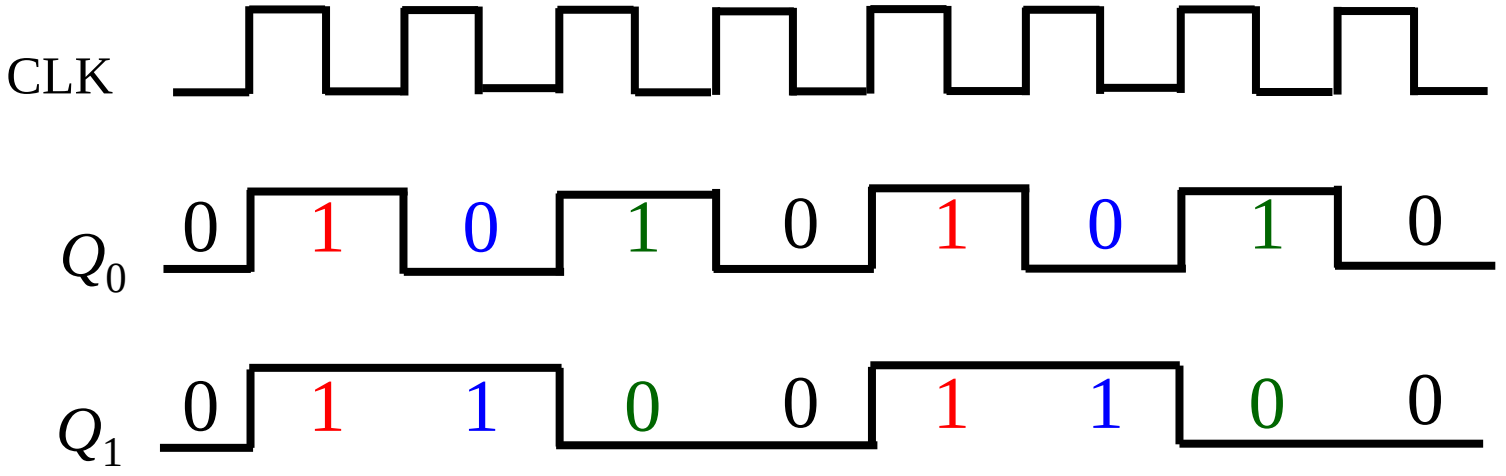
两位二进制加法计数器

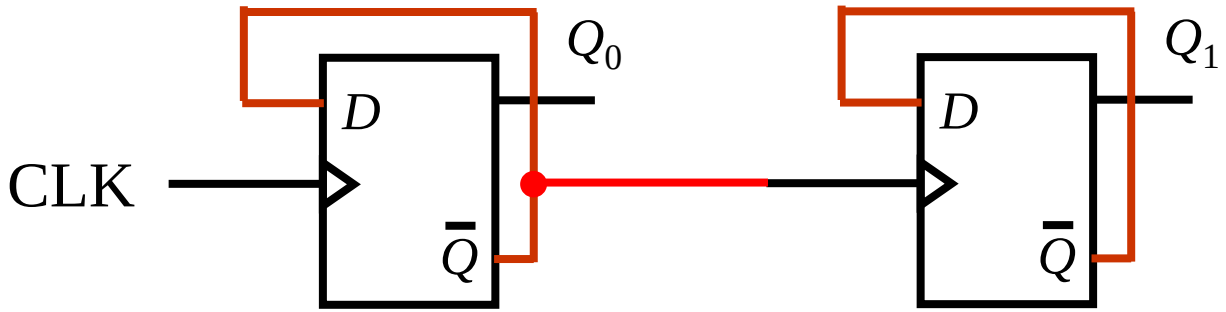

CLK
Q_0
Q_1
0 1 0 1 0 1 0 1 0
0 0 1 1 0 0 1 1 0

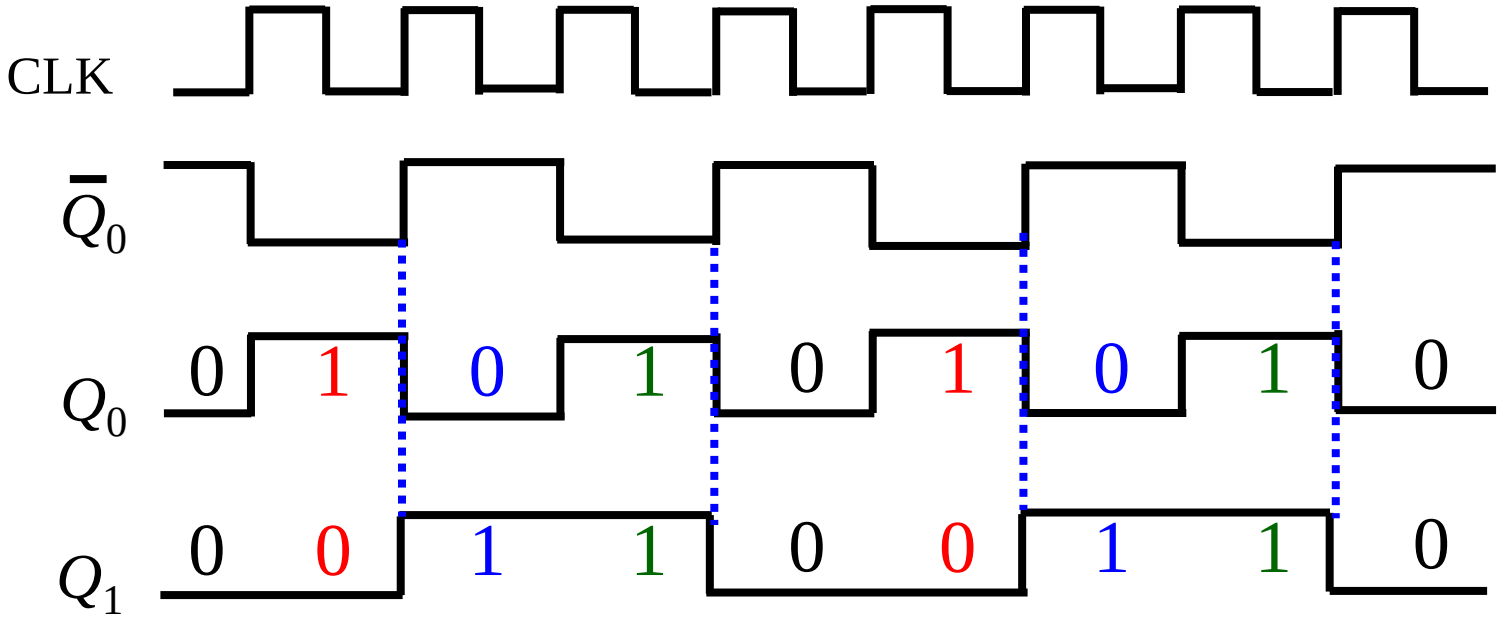

Q_1Q_0
00
01
11
10



$$Q^{n+1}=D=\bar{Q}^{n}$$




CLK
D
Q_0
$\bar{Q}$
D
Q_1
$\bar{Q}$


CLK
$\bar{Q}_0$
Q_0
0 1 0 1 0 1 0 1 0
Q_1
0 0 1 1 0 0 1 1 0